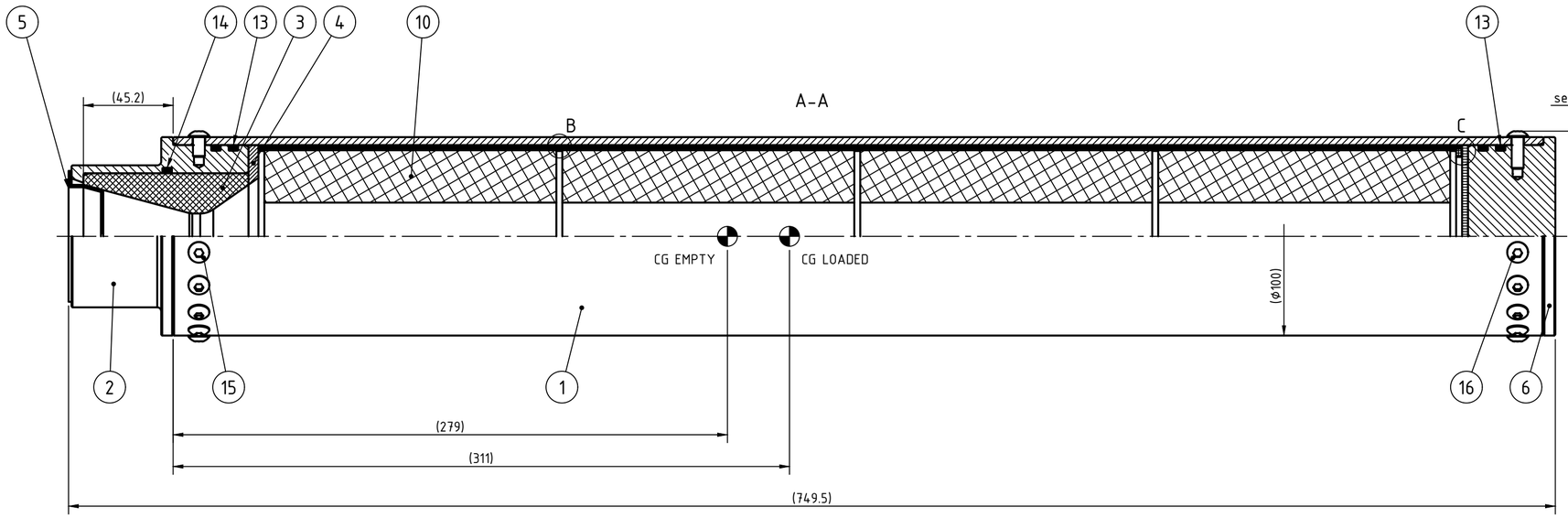
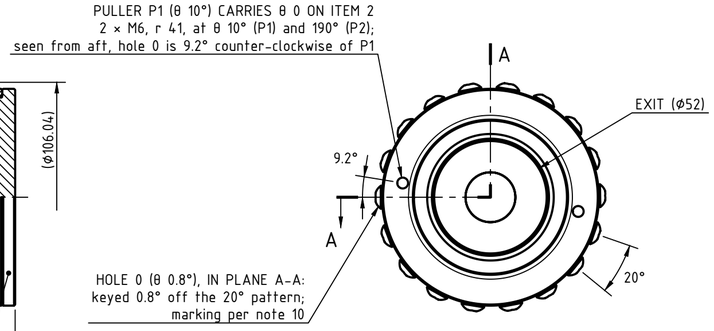
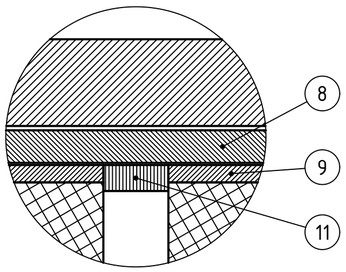




AFT END VIEW (item 5 not shown)



B (5:1)

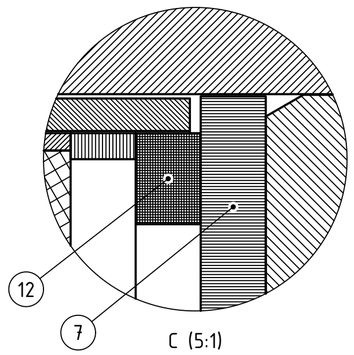
SECTION HATCHING (ISO 128-50)

- Metal, alternating: 1 (45°); 2, 6 (135°)
- Graphite: 3
- Glass phenolic: 4
- Paper phenolic: 8 (detail B)
- Spiral-wound kraft: 9 (detail B)
- Linen phenolic: 11 (detail B)
- Propellant, APCP: 10
- EPDM: 7
- Silicone sponge: 12 (detail C)
- Solid: 13, 14 (O-rings), 5; at 1:2 also 8, 9, 11
- Not sectioned: screws 15, 16

MOTOR DATA

PARAMETER	VALUE (nominal run unless stated)
Propellant	APCP 70/16/14 (AP/Al/HTPB)
Density, as cast	1 718 kg/m³ (98 % of 1 753 theoretical)
Grains	4 × BATES, ø86.8 / ø34 × 147.2 both ends burning
Designation	N2203
Total impulse / class	10 495 N·s / N (2.5 % into N) M/N borderline until a static fire
Thrust, average / peak	2203 / 2383 N
Burn time	4.76 s
Specific impulse	211.2 s
Pc, average / peak	3.77 / 4.05 MPa
Kn, initial / peak	247.8 / 281.1, nominal simulated run (erosive) 279.5 max geometric (uniform regression)
MEOP	7.0 MPa
Hydrostatic proof	10.5 MPa (1.5 × MEOP), SRM100-001
Throat / exit	ø23.00 / ø52.0
Expansion ratio	5.11
Mass, loaded / empty	10.601 / 5.534 kg
Propellant mass	5.067 kg (fraction 0.478)
CG, loaded / empty	311 / 279 mm fwd of the case aft face
Bound run (sizing)	N2523: 11 504 N·s, Pc peak 4.59 MPa (the paper's ideal inputs)

– Nominal run: 1 718 kg/m³ as cast, c* × 0.95, Mukunda-Paul erosive burning, grain at 20 °C. Predictions from srm_ballistics, not measured: the static fire settles the class.



C (5:1)

NOTES

1. Assemble per sheet 2 (aft end: items 2–5, 13–15) and sheet 3 (liner, grain stack and forward closure).
2. Hydrostatic proof per SRM100-001 before the first firing and after any repair, any retired-and-replaced part, or a temperature label reading ≥ 232 °C: 3.5 / 7 / 10.5 MPa, 300 s holds, relief valve 11 MPa, decay ≤ 2 %, permanent set ≤ 0.02 mm. The proof build uses proof dummy SRM100-120 in place of item 3 (item 14 stays); items 4, 5 and 7–12 are not fitted. Never wet or hydro-test the graphite.
3. ENERGETIC MATERIAL: item 10 (APCP) makes this an N-class research motor. Make, store, transport and fire it only as local law allows and under the Tripoli Research (Level 3) and NFPA 1127 rules; minimum distance for N: 1000 ft (≈ 305 m). The site RSO's rules govern.
4. Nozzle cap item 5: fitted at assembly as the debris seal, taped per SRM100-106 note 6, kept on through storage and transport. At the stand it comes off only to insert the igniter (leads shunted, out of its ø3 hole), then it is refitted and re-taped with new tape. It blows out at ignition.
5. The igniter is not part of this drawing (README: 3 g BKNO3 or pyrogen, tube ≤ ø10 × 40, dual-bridge e-match). It goes in last, at the stand, leads shunted. Never move the motor with an igniter in it.
6. Screws, new ISO 7380-1 010.9 every firing: 18 × item 15 at 3–3.5 N·m, nickel anti-seize on the threads only, head face dry; 18 × item 16 at 4–6 N·m with Loctite 243. Tighten criss-cross (opposite pairs) in 3 passes, sheets 2 and 3. Never interchange the lengths: an M6×16 in item 2 bottoms with its head proud, an M6×10 in item 6 engages only 1 d.
7. O-rings 13, 14: FKM 90A (Parker V0709-90 or equal), new every firing. Lube them, the liner OD and the insulator edges with Parker Super O-Lube or DC-55, thin film. Bond item 7 to item 6 with RTV106 or equal on SS4004P primer, bondline 0.1–0.2 (sheet 3).
8. Seal zones of item 1, both ends: each honed to the smallest diameter that cleans it up, A (the smallest reading in the zone, recorded per zone, SRM100-101 note 4), 92.000 ≤ A ≤ 92.070, held to A +0.030/0, Ra 0.4 (0.8 max); the bore is ≤ 92.10 everywhere. Items 2 and 6 are cut to their own case zone's A: land A – 0.06 0/–0.02, groove bottom A – 5.67 0/–0.04.
9. Each case end and its closure are a match-drilled, serialised pair. θ 0 is carried on item 1 by hole 0, the scribed clocking line and the AFT / FWD engraving; on items 2 and 6 by puller P1 at θ 10° (r 41 on item 2, r 37.5 on item 6). Hole 0 (θ 0.8°) is 9.2° clockwise of P1 seen from forward and 9.2° counter-clockwise seen from aft (end view); keyed 0.8° off the 20° pattern, a pair goes together one way only.
10. Marking and temperature labels per SRM100-101 note 8 (item 1) and SRM100-102 note 8 (item 2); marking of item 6 per SRM100-103 note 11. Labels: irreversible 177 / 204 / 232 / 260 °C strips, new for every firing, at θ 90° and 270° on the item 1 OD at z 20–35 and the item 2 aft body ø71.58, clear of the cap tape. After firing read and record them before extraction; ≥ 232 °C: retire or re-proof that end (note 2).
11. No pressure port: item 6 is solid (no port, no pocket) and item 7 is a full disc. The motor is fired without a chamber transducer; the static-fire record is the load cell. A transducer needs a new, ported item 6 and a re-proof. The proof fills and bleeds through SRM100-120's port (SRM100-001).
12. Liner item 8: cut to the measured cavity minus 0.5 float. Check the float by depth gauging before item 6 goes in, not with a feeler (the gap is closed off). Take-up ring item 12: 3.2, 4.8 or 6.4 sheet chosen from the measured gap for 25–50 % compression. Both per sheet 3.

NOTES (continued)

13. After every firing replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels; re-use items 1, 2, 6 after inspection (SRM100-101 R10 for the case), and item 3 while the throat is ≤ ø23.9 and uncracked.
14. Static fire: item 6's forward face is the thrust face, flat from the axis to ø100, broken only by the 2 M6 puller holes at r 37.5; a load cell or button bears anywhere clear of them. Flight: the airframe needs a ø108 keep-out for the screw heads and a thrust ring or a forward-closure thrust path.
15. All dimensions on this sheet are reference dimensions. Masses and CG are from the STEP solids (mass_budget.json); CG measured forward from the case's aft end face.
16. Tooling, referenced by name only: acetal hole plugs, push sleeve, casting mandrel, puller bridge with a split clamp collar (100.0 bore) on the case OD ≥ 50 from the end being extracted, clear of the screw heads 13 from each end face (ø106.04 sweep), the labels and the marking 20–45 from each end face.
17. Proof dummy SRM100-120 and proof assembly SRM100-001 are proof hardware only: never fired, not part of this assembly.

USED ON: HELIOS rocket – SRM-100 motor (static-fire stand, then flight)

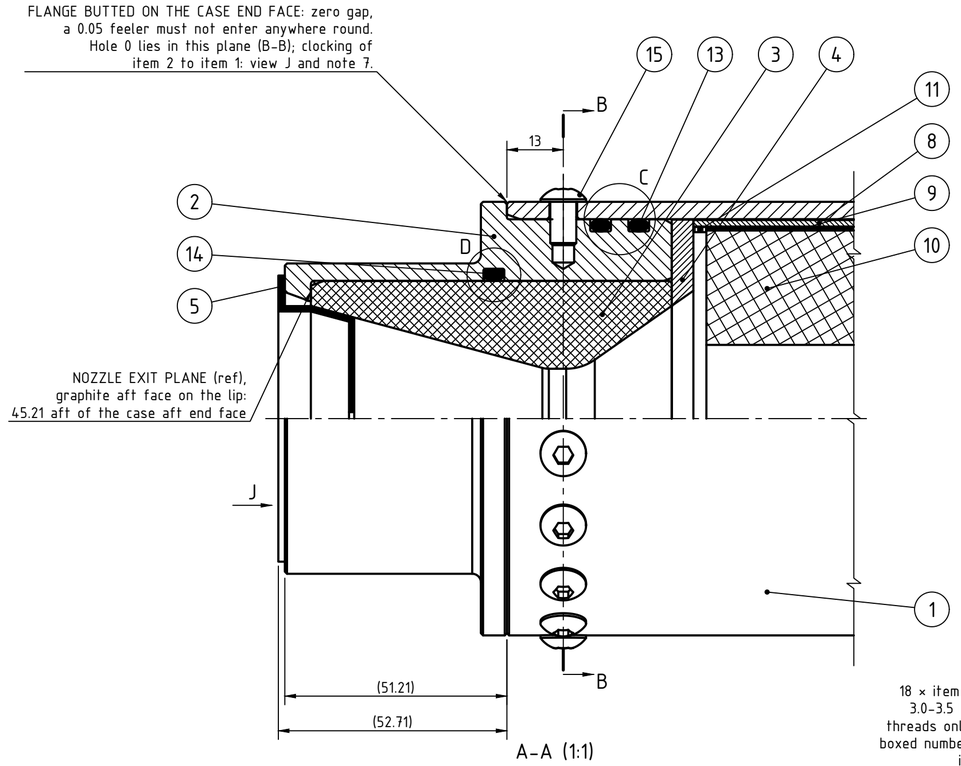
ITEM	QTY	PART NO. / STANDARD	DESCRIPTION	MATERIAL	MASS g (each)
–	4	Temp. label 177–260 °C	On 1, 2: new each firing (note 10)	irreversible strip	–
–	a/r	PVC tape, 19 wide	Cap tape: 5 (SRM100-106 note 6)	PVC electrical tape	–
–	a/r	RTV106 + SS4004P	Bond 7 to 6, primed (note 7)	RTV silicone + primer	–
–	a/r	Loctite 243; anti-seize	Threads: 16 (243), 15 (anti-seize)	anaerobic / nickel	–
–	a/r	Super O-Lube or DC-55	Lube: 13, 14, liner OD (note 7)	silicone, thin film	–
16	18	ISO 7380-1 M6×16-010.9	Button-head screw, fwd	steel 010.9	4.7
15	18	ISO 7380-1 M6×10-010.9	Button-head screw, aft	steel 010.9	3.4
14	1	AS568-230	O-ring, insert seal	FKM 90A	3.8
13	4	AS568-237	O-ring, case seal	FKM 90A	5.1
12	1	SRM100-108	Take-up ring	silicone sponge	1.7
11	5	SRM100-109	Grain spacer	linen phenolic	1.3
10	4	SRM100-112	Propellant grain, BATES	APCP	1266.8
9	4	SRM100-111	Casting tube	spiral-wound kraft	30.8
8	1	SRM100-110	Liner	paper phenolic	347.5
7	1	SRM100-107	Forward insulator	EPDM	22.8
6	1	SRM100-103	Forward closure	6061-T651	770.5
5	1	SRM100-106	Nozzle cap	LDPE	6.6
4	1	SRM100-105	Nozzle insulator	glass phenolic	38.8
3	1	SRM100-104	Nozzle insert	isomolded graphite	311.1
2	1	SRM100-102	Nozzle holder (aft closure)	304 stainless	1501.2
1	1	SRM100-101	Motor case	6061-T6 drawn tube	2234.4

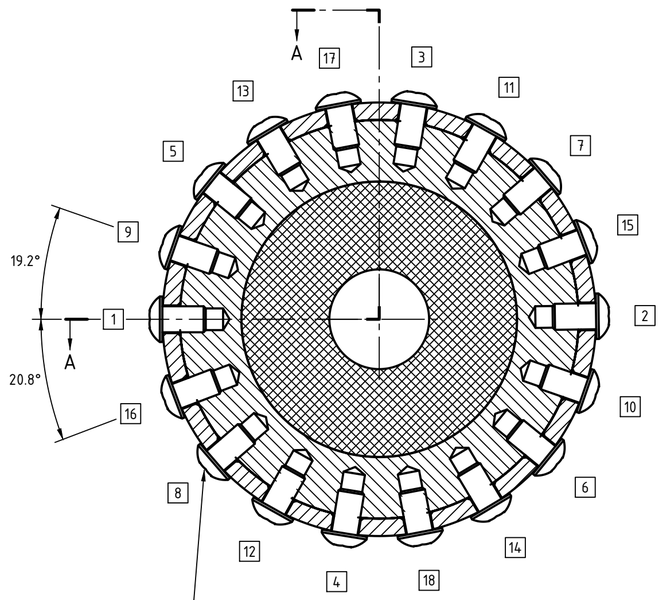
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
See parts list	10.60 kg	ISO 2768-mK	–	–

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
ASSEMBLY DRAWING	NOT CHECKED – FOR REVIEW	1:2		mm

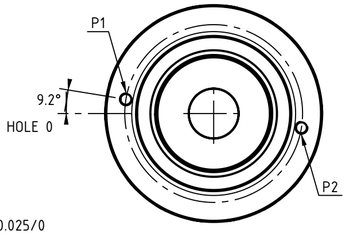
LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	GENERAL ARRANGEMENT SRM-100 motor, loaded – parts list, motor data

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-000	A	1 / 3

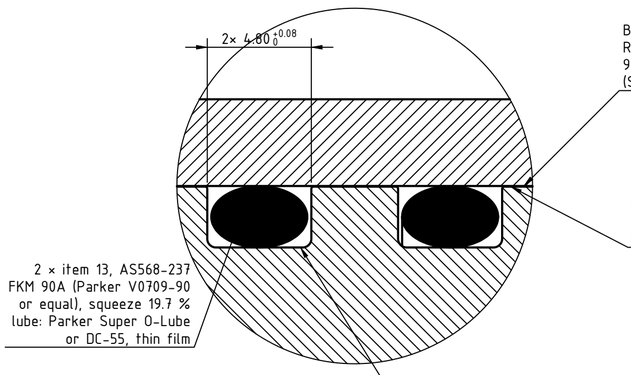




18 × item 15, M6×10–010.9 (ISO 7380-1) 3.0–3.5 N·m, nickel anti-seize on the threads only, head face dry. Order: the boxed numbers, 3 passes (note 8). Holes in item 1: ⌀6.40 +0.03/0 (ref)



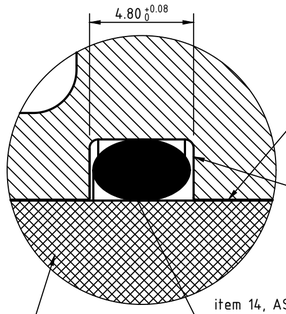
At entry, seen from aft: hole 0 of item 2 on hole 0 of item 1, clocking line straight across both, AFT markings side by side (note 7). Hole 0 is 9.2° counter-clockwise of puller P1 (θ 10°); P2 at θ 190°, both on r 41: 2 × M6×1–6H ↧8, ⌀5 ↧10 (SRM100-102).



BORE, item 1: ⌀A +0.030/0, honed Ra 0.4 (0.8 max), circumferential lay; 92.000 ≤ A ≤ 92.070, recorded per zone (SRM100-101). Drawn at A = 92.000.

LAND, item 2: ⌀A – 0.06 0/–0.02 (⌀91.94 at A = 92.000) clearance on ⌀ 0.06–0.11 (0.030–0.055 radial)

2 GROOVES ⌀A – 5.67 0/–0.04 (⌀86.33 at A = 92.000) gland depth 2.835 (ref)



BORE, item 2: ⌀63.58 +0.025/0
INSERT, item 3: ⌀63.50 0/–0.05
clearance on ⌀ 0.080–0.155

GROOVE ⌀69.14 +0.05/0 at z –3 gland depth 2.820 (ref)

ITEM 3: slide in by hand, aft end first. Never press or knock it in. Push sleeve: extraction only (note 13).

item 14, AS568-230, FKM 90A (Parker V0709-90 or equal), squeeze 20.1 %, lube as item 13, fitted in item 2's groove first

AXIAL STATIONS, z from the case aft end face, + forward

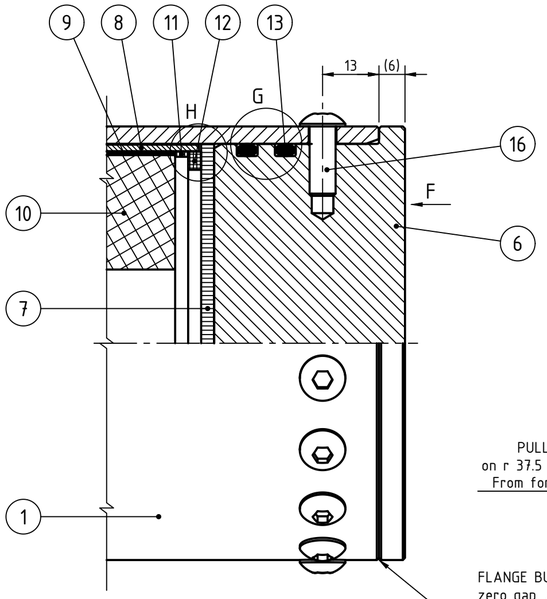
STATION (ref)	z
Cap flange back face (motor aft end)	–52.71
Holder aft face	–51.21
Nozzle exit plane (graphite aft face)	–45.21
Holder flange aft face	–6.00
Insert O-ring groove, centre	–3.00
Case aft end face = flange face (datum)	0.00
Bolt row, 18 × item 15	13.00
Case O-ring grooves, centres	21.60 / 30.40
Holder and insert forward faces	38.00
Insulator fwd face = liner and spacer 0 aft	43.00
Grain 1 aft face	46.00

AFT END BUILD-UP AND INSTALLATION

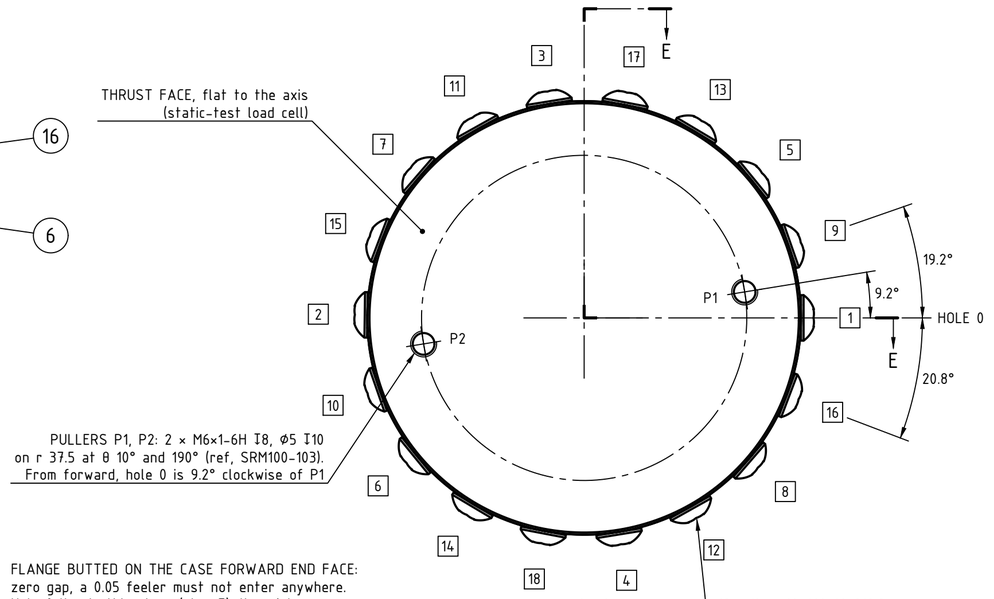
1. Item numbers, materials and part numbers: parts list on sheet 1. Tooling named here (hole plugs, push sleeve, puller bridge) is not part of this drawing set.
2. Before the first build (README, case): item 1 per SRM100-101, each seal zone honed to the smallest diameter that cleans up, A (the smallest reading in the zone, 92.000 ≤ A ≤ 92.070), held to A +0.030/0 and recorded per zone; items 2 and 6 cut to their zone's A; each end match-drilled with its closure, serialised and marked (SRM100-101 note 8, SRM100-102 note 8, SRM100-103 note 11), clocking line scribed; hole baseline recorded (SRM100-101 R7). Then the hydrostatic proof (SRM100-001, proof dummy SRM100-120 in place of item 3, no insulators, liner or propellant): before the first firing and after any repair, any retired-and-replaced part or a label reading ≥ 232 °C (note 12).
3. The parts replaced after every firing (note 14) are always new. Lubricant for every O-ring, the liner OD and the insulator edges: Parker Super O-Lube or DC-55, thin film.
4. Fit item 14 in the bore groove of item 2, then 2 × item 13 in the grooves of its seal land (details C, D). Item 2 is cut to the aft-zone bore A recorded on SRM100-101 (92.000 ≤ A ≤ 92.070): land A – 0.06 0/–0.02, groove bottoms A – 5.67 0/–0.04 (91.94 / 86.33 at A = 92.000). Details C and D are drawn at A = 92.000.
5. Slide insert item 3 into item 2 by hand, aft (exit) end first, until it seats on the lip. Never press or knock it. Check: its forward face lies flush with item 2's forward face to 0.10 below it (seat 83.21 +0.05/0, insert 83.21 0/–0.05). Then seat nozzle insulator item 4 on item 2's forward face.
6. Hole plugs: push 18 flat-ended acetal (Delrin) plugs, ⌀6.35 × 3.85 with an outside head, into the case's aft-end holes. Their ends sit about 0.15 mm shy of the bore, so no O-ring is dragged across an open hole edge.
7. Clocking (view J): enter item 2 with its hole 0 on the case's hole 0, the scribed clocking line straight across the case end and the flange and the AFT markings of items 1 and 2 (SRM100-101 note 8, SRM100-102 note 8) side by side. Check: seen from aft, hole 0 is 9.2° counter-clockwise of puller P1 (θ 10°). Push in by hand until the flange butts on the case end face (zero gap).
8. Pull each plug as its screw goes in. Screws 18 × item 15, 3.0–3.5 N·m, nickel anti-seize on the threads only, head face dry. Tighten in the boxed order of section B–B (opposite pairs, criss-cross): pass 1 to about 1.5 N·m, pass 2 to final torque, pass 3 at final torque hole by hole round the circle, as a check. Never fit an M6×16 (item 16) here: it bottoms with its head proud.
9. Load the liner and grain stack and close the forward end: sheet 3.
10. Temperature labels, new for every firing: irreversible 177 / 204 / 232 / 260 °C strips on item 1's OD at z 20–35, θ 90° and 270° (SRM100-101 note 8), and on item 2's aft body ⌀71.58 at θ 90° and 270° (SRM100-102 note 8), clear of the cap tape (note 11).
11. Cap item 5: fit it now, in the exit, flange on item 2's aft face, taped per SRM100-106 note 6. It is the debris seal and stays on through storage and transport. At the stand it comes off only to insert the igniter, which always goes in last (leads shunted, out through the cap's ⌀3 hole); then it is refitted and re-taped with new tape (sheet 1 note 4). Never move the motor with an igniter in it.
12. After every firing, before extraction: read and record the temperature labels on items 1 and 2 (note 10). A reading ≥ 232 °C: retire or re-proof that end (SRM100-101 R10, SRM100-001).
13. Extraction, never lever or knock. Closure: split clamp collar (100.0 bore) on the case OD at least 50 mm from the end, clear of the screw heads at z 13 (sweep ⌀106.04 ref), the labels and the marking at z 20–45; puller bridge and 2 × M6 draw screws in item 2's puller threads P1, P2 (view J). Insert: with item 2 out, support it on its forward face in an arbor press and push item 3 out forward with the push sleeve (⌀52.4 ID × ⌀53.8 OD) through the lip bore onto its push face, r 26.2–27.0. The push sleeve is for extraction only.
14. Then, after every firing: measure the throat. Replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels. Re-use items 1, 2 and 6 after inspection (SRM100-101 R10 for the case), and item 3 while its throat is ≤ ⌀23.9 and it is uncracked. On the aft case end, measure hardness and conductivity again and re-gauge every bolt hole against its baseline (SRM100-101 R10).

USED ON: SRM-100 motor (Helios) – static test and flight

MATERIAL: See parts list, sheet 1	MASS: 10.60 kg	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: –	EDGES: –
DOCUMENT TYPE: ASSEMBLY DRAWING	DOCUMENT STATUS: NOT CHECKED – FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: MOTOR ASSEMBLY – AFT END — Aft closure build-up and installation, seal glands, tightening sequence			
CREATED BY: text-to-cad / Claude	APPROVED BY: –	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-000	REV: A
				SHEET: 2 / 3



E–E (1:1)



PULLERS P1, P2: 2 × M6×1-6H ↧8, ⌀5 ↧10 on r 37.5 at θ 10° and 190° (ref, SRM100-103). From forward, hole 0 is 9.2° clockwise of P1

FLANGE BUTTED ON THE CASE FORWARD END FACE: zero gap, a 0.05 feeler must not enter anywhere. Hole 0 lies in this plane (view F). Item 6 is a serialised pair with item 1; clocking: note 12.

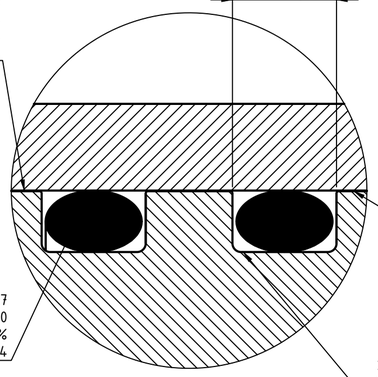
F (1:1)

18 × item 16, M6×16-010.9 (ISO 7380-1), 4.0–6.0 N·m, Loctite 243. Order: the boxed numbers, 3 passes (note 13). Holes in item 1: ⌀6.40 +0.03/0; 12 engaged in item 6 (ref)

LOADING AND FORWARD END CLOSING

1. Item numbers and materials: parts list, sheet 1. Tooling (casting mandrel, hole plugs, clamp collar, puller bridge) is not in this drawing set.
2. Aft end first (sheet 2); first build and proof: sheet 2, note 2. Bond item 7 to item 6 only after the proof (run with no insulators or liner).
3. Bond item 7 to item 6's inboard face (README assembly step 1, SRM100-107): abrade, IPA wipe, silicone primer SS4004P, RTV106 or equal over the whole face, bondline 0.1–0.2, OD centred on the land within 0.1; cure 24 h. Then measure t, flange bearing face to item 7's aft face, at 4 points, mean (nominal 41.00 + bondline).
4. Lube (O-rings, liner OD, insulator edges): Parker Super O-Lube or DC-55, thin film. Parts replaced per note 17 (labels included) are always new.
5. Grains are cast in their casting tubes (item 9) on the casting mandrel; the tube stays on the grain. Mandrel out, measure and record every grain length (147.20 ±0.10) and spacer (3.00 ±0.02): stack-up table.
6. Cavity and liner: with the aft end closed, measure the depth d_N from the case forward end face to item 4's forward face (nominal 647.80). Cavity C = d_N − t (nominal 606.80). Cut item 8 to C − 0.50 0/−0.10 (SRM100-110), lubricate its OD (note 4) and slide it in by hand until it bears on item 4.
7. Load by hand, aft to forward: S0 on item 4, G1, S1, G2, S2, G3, S3, G4, S4 (stack view E–E). Each tube and spacer must slide freely in the liner.
8. Liner float, in place of the README feeler check (the gap is closed off once item 6 is in): depth d_L from the case forward end face to the liner's forward end ≥ t + 0.30 at 4 points (nominal 41.50 for t = 41.00).
9. Take-up gap: depth d_S from the case forward end face to the top of S4 at 4 points, mean; g = d_S − t (nominal 44.00 − 41.00 = 3.00; ±0.50 stack, ±0.10 case length). Measure the free thickness t_s of item 12 (4 points, mean) and accept 0.50 ≤ g / t_s ≤ 0.75 (25–50 % compression; nominal 37.5 %; SRM100-108): sheet 3.2 for g 1.60–2.40, 4.8 for g 2.40–3.60, 6.4 for g 3.20–4.80. Lay item 12 on S4.
10. Hole plugs: push 18 acetal plugs into the case's forward-end holes before item 6 goes in (as sheet 2, note 6).
11. Fit 2 × item 13, FKM 90A (Parker V0709-90 or equal), in the grooves of item 6 (detail G). Item 6 is cut to the forward-zone bore A recorded on SRM100-101 (92.000 ≤ A ≤ 92.070): land A − 0.06 0/−0.02, groove bottoms A − 5.67 0/−0.04. Detail G is drawn at A = 92.000.
12. Clocking (view F): enter item 6 with hole 0 on the case's hole 0, the scribed clocking line straight across case end and flange, the FWD markings of items 1 and 6 (SRM100-101 note 8, SRM100-103 note 11) side by side. Check: seen from forward, hole 0 is 9.2° clockwise of puller P1 (θ 10°). Serialised pair: never fit another closure. Push in by hand until the flange butts on the case end face (zero gap).
13. Pull each plug as its screw goes in. Screws 18 × item 16 with Loctite 243, in the boxed order of view F (opposite pairs, criss-cross): pass 1 to about 2.0 N·m, pass 2 to 4.0–6.0 N·m, pass 3 at final torque hole by hole round the circle, as a check. Never an M6×10 (item 15) here: 1 d engagement only.
14. No pressure port (sheet 1 note 11): item 6 is solid (no port, no pocket) and item 7 is a full disc; nothing is fitted or plugged at the forward end.
15. Igniter last, at the stand; cap per sheet 2 note 11. It goes up the throat about 615 mm from the nozzle exit, to the middle of G4's core (z 570.2).
16. Extraction, labels read first (sheet 2 note 12), never lever or knock: screws out; split clamp collar (100.0 bore) on the case OD at least 50 mm from the forward end face, clear of the screw heads at 13 (sweep ⌀106.04 ref) and the marking at 20–45 from that face; puller bridge, 2 × M6 draw screws in item 6's P1, P2 (view F), or react on the stand's thrust plate.
17. After every firing: replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels. Re-use items 1, 2 and 6 after inspection (SRM100-101 R10 for the case), and item 3 while its throat is ≤ ⌀23.9 and it is uncracked. Scrape item 7 and the old RTV off item 6 with a plastic scraper (SRM100-103); re-gauge every bolt hole against its baseline (SRM100-101 R10).

BORE, item 1: ⌀A +0.030/0, honed Ra 0.4 (0.8 max), circumferential lay; 92.000 ≤ A ≤ 92.070, recorded per zone (SRM100-101). Drawn at A = 92.000.



LAND, item 6: ⌀A − 0.06 0/−0.02 (⌀91.94 at A = 92.000) clearance on ⌀ 0.06–0.11 (0.030–0.055 radial)

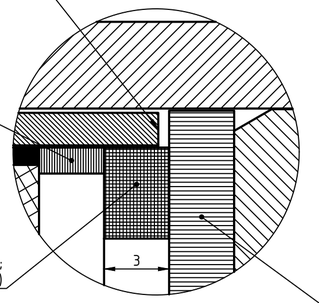
2 × item 13, AS568-237 FKM 90A (Parker V0709-90 or equal), squeeze 19.7 % lube: note 4

2 GROOVES ⌀A − 5.67 0/−0.04 (⌀86.33 at A = 92.000) gland depth 2.835 (ref)

G (5:1)

LINER END, item 8: floats 0.50 short of item 7 (0.30 min, note 8)

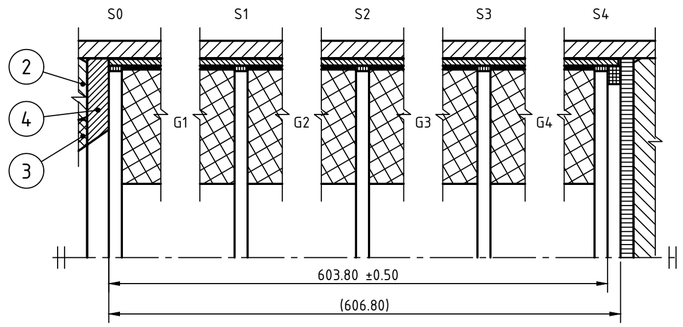
S4, item 11, 3.00 ±0.02



item 12, 4.80 free, 3.00 installed (37.5 %); 27.1–47.9 % over the ±0.50 stack (note 9)

item 7, EPDM 3.0 ±0.2, RTV-bonded to item 6 (note 3); the 0.1–0.2 bondline is not in the model

H (5:1)



E–E (1:1), GRAIN STACK – INTERRUPTED VIEW

GRAIN STACK AXIAL STACK-UP (nominal; z from the case aft end face)

POS	ELEMENT	ITEM	LENGTH	TOL	z AFT	z FWD
S0	Spacer 0 (on item 4)	11	3.00	±0.02	43.00	46.00
G1	Grain 1 in its casting tube	10, 9	147.20	±0.10	46.00	193.20
S1	Spacer 1	11	3.00	±0.02	193.20	196.20
G2	Grain 2 in its casting tube	10, 9	147.20	±0.10	196.20	343.40
S2	Spacer 2	11	3.00	±0.02	343.40	346.40
G3	Grain 3 in its casting tube	10, 9	147.20	±0.10	346.40	493.60
S3	Spacer 3	11	3.00	±0.02	493.60	496.60
G4	Grain 4 in its casting tube	10, 9	147.20	±0.10	496.60	643.80
S4	Spacer 4	11	3.00	±0.02	643.80	646.80
STACK	5 spacers + 4 grains (worst case)	–	603.80	±0.50	43.00	646.80
T	Take-up ring, installed (4.80 free)	12	3.00	= g	646.80	649.80
C	Cavity = stack + take-up	–	(606.80)	meas.	43.00	649.80
L	Liner, cut to C − 0.50	8	606.30	0/−0.10	43.00	649.30
–	Liner float to item 7	–	0.50	0.30 min	649.30	649.80
I	Forward insulator, EPDM, bonded	7	3.00	±0.20	649.80	652.80
P	Forward closure, plug	6	38.00	gen.	652.80	690.80
F	Forward closure, flange	6	6.00	gen.	690.80	696.80
–	Case (aft end face z 0)	1	690.80	±0.10	0.00	690.80

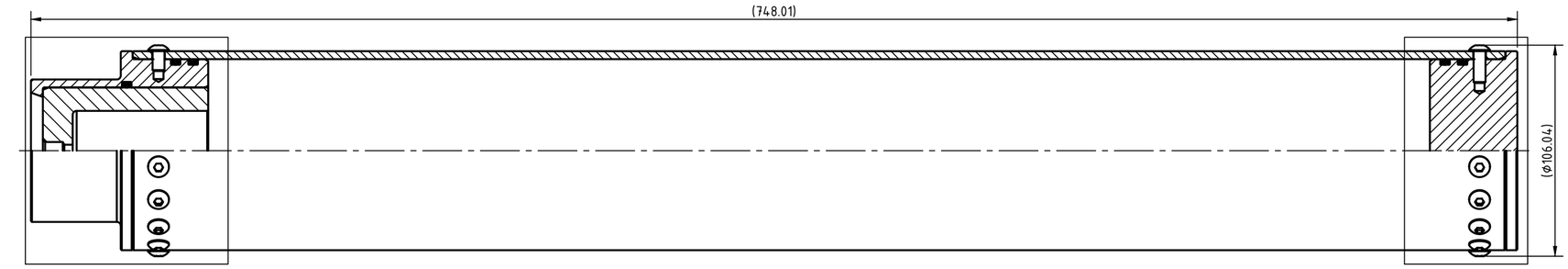
Stations (ref): O-ring grooves 660.40 / 669.20, bolt row 677.80, thrust face 696.80.
Motor overall 749.51 (cap back face z −52.71 to the thrust face). Igniter tip station z 570.2 (note 15).
TOL gen. = ISO 2768-mK (SRM100-103), case per SRM100-101; the build uses the measured t, d_N and d_S.

GAUGING RECORD AT LOADING (one per firing, file with the motor log)

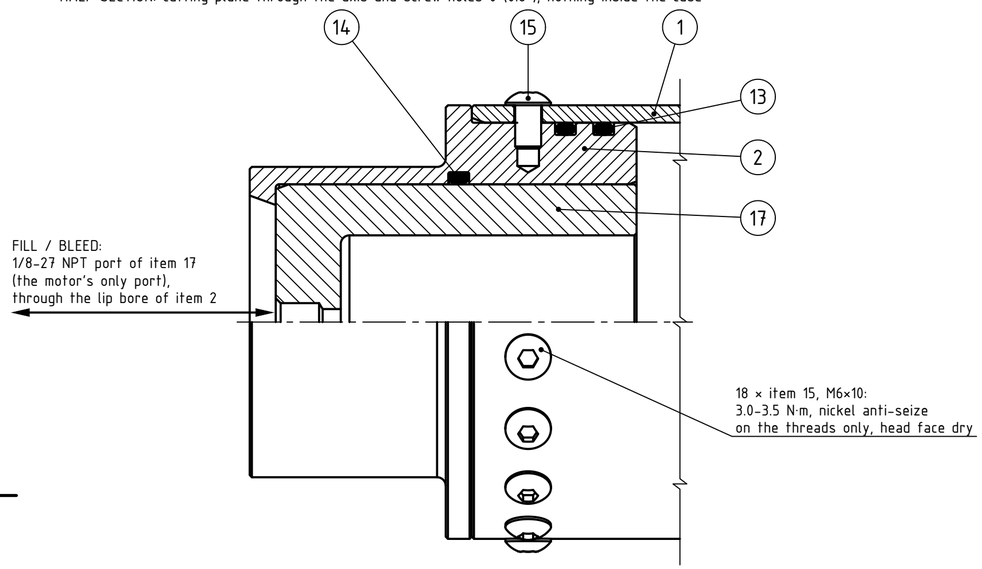
QUANTITY (notes 3–9, 12, 13)	NOMINAL	ACCEPT	MEASURED
t: item 6 flange face to item 7, 4 pts	41.00	+ bondline	
d_N: case end face to item 4, 4 pts	647.80	record	
C = d_N − t: cavity	606.80	record	
L: liner cut length	606.30	C − 0.50 0/−0.10	
Stack: measured spacers + grains	603.80	±0.50	
d_L: case end face to liner end, 4 pts	41.50	≥ t + 0.30	
d_S: case end face to S4 top, 4 pts	44.00	record	
g = d_S − t: take-up gap	3.00	record	
t_s: item 12 free thickness, 4 pts	4.80	sheet per g	
g / t_s	0.625	0.50–0.75	
Flange gap to the case end face	0	0.05 feeler: no entry	
Screw torque, final pass	4.0–6.0 N·m	3 passes	

USED ON: SRM-100 motor (Helios) – static test and flight

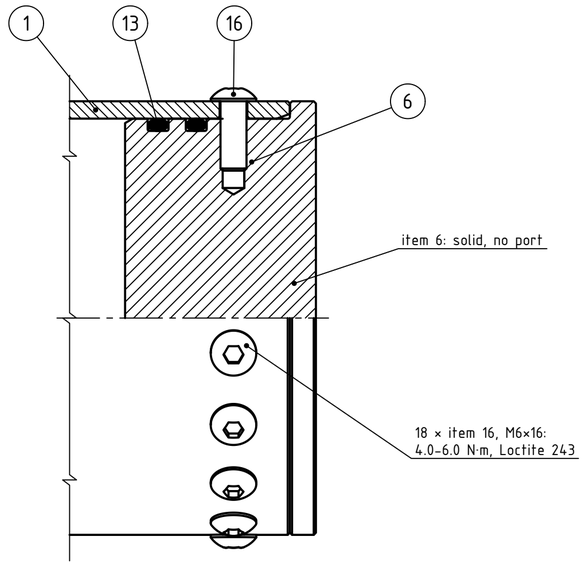
MATERIAL: See parts list, sheet 1	MASS: 10.60 kg	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: –	EDGES: –
DOCUMENT TYPE: ASSEMBLY DRAWING	DOCUMENT STATUS: NOT CHECKED – FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: MOTOR ASSEMBLY – FORWARD END — Forward closure build-up, grain stack loading and stack-up, tightening			
CREATED BY: text-to-cad / Claude	APPROVED BY: –	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-000	REV: A — SHEET: 3 / 3



B HALF SECTION: cutting plane through the axis and screw holes 0 (0.8°); nothing inside the case C



B (1:1)



C (1:1)

PROOF BUILD

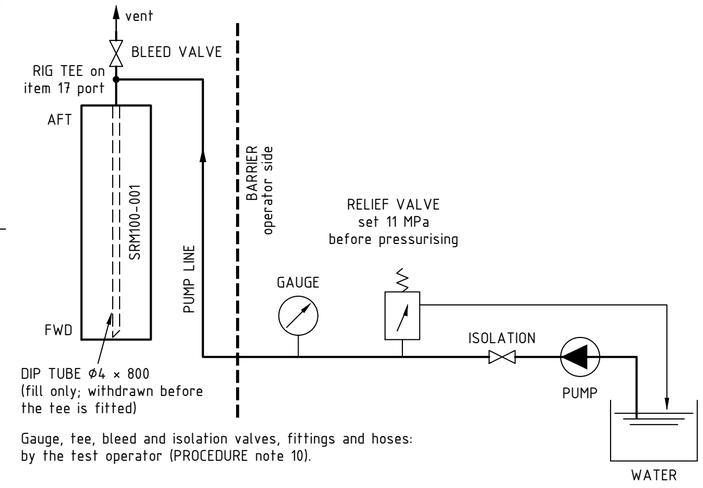
1. Build list: the flight case item 1, both closures (items 2, 6), all five O-rings (4 × item 13, item 14) and all 36 screws (items 15, 16), with proof dummy item 17 in the insert's gland. No graphite insert item 3 (it must stay dry; a cracked one is scrap), no insulators, liner, grain or cap: nothing inside the case.
2. Each closure goes to its own case end: match-drilled, serialised pair. θ0: on the case, hole 0, the clocking line scribed across the pair and the AFT / FWD marking; on each closure, puller P1 at θ 10° (item 2 r 41, item 6 r 37.5). Hole 0 is at 0.8°, 9.2° clockwise of P1 seen from forward (counter-clockwise seen from aft).
3. O-rings items 13, 14: FKM 90A, Parker V0709-90 or equal. Lube: Parker Super O-Lube or DC-55, thin film. Fit item 14 in item 2's bore groove, then 2 × item 13 in each closure's land grooves. Items 2 and 6 are cut to their own case end's bore A (fits table).
4. Slide item 17 into item 2 by hand, aft end first, until it seats on the lip. Never press or knock it (view B). Check: its forward face lies flush with item 2's forward face to 0.10 below it.
5. Hole plugs (tooling): before each closure goes in, push 18 flat-ended acetal plugs, ϕ6.35 × 3.85 with an outside head, into that end's holes; their ends sit about 0.15 mm shy of the bore, so no O-ring is dragged across an open hole edge. Pull each plug as its screw goes in.
6. Enter each closure at its own end (item 2 AFT, item 6 FWD) with its hole 0 on the case's hole 0 and clocking line, until its flange butts on the case end face: zero gap.
7. Torques as for flight. Aft: 18 × item 15 into item 2 (304), 3.0–3.5 N·m, nickel anti-seize on the threads only, head face dry. Forward: 18 × item 16 into item 6 (6061), 4.0–6.0 N·m, Loctite 243. Criss-cross, in the boxed order of SRM100-000 sheet 2 (aft) / sheet 3 (fwd), 3 passes: about half the minimum torque (1.5 / 2.0 N·m), then to torque, then a check pass hole by hole round the circle. Do not mix the lengths: an M6×16 in the holder bottoms with its head proud; an M6×10 in the forward closure engages only 1 d.
8. Before the first proof, record the hole baseline of all 36 case holes (SRM100-101 record R7), keyed to the serialised pair: the axial length of each hole, and the ligament from the case end face to the hole edge (nominal 13.0 − 3.2 = 9.80). Each hole is judged against its own baseline only, never an absolute ligament limit (SRM100-101 R7).
9. Mark the case OD stations S1 … S5 with removable tape and record the OD at each before the test: stations, directions and micrometer per SRM100-101 record R9.
10. Disassembly, never lever or knock: split clamp collar (100.0 bore) on the case OD at least 50 mm from the end being extracted (both ends), clear of the screw heads at z 13 (ϕ106.04 sweep, ref), the labels and the marking at z 20–45 (z from that end face); bridge and 2 × M6 draw screws in the closure's puller threads, 2 × M6×1-6H ↧8 / ⌀5 ↧10 at θ 10° and 190° (item 2 r 41, item 6 r 37.5). Item 17 comes out of item 2 as the insert does: holder on its forward face in an arbor press, push sleeve through the lip bore.
11. USED ON: SRM-100 motor (Helios) – proof before the first firing and after any repair.

HYDROSTATIC PROOF – PROCEDURE AND ACCEPTANCE

1. Medium: water. Proof pressure 10.5 MPa = 1.5 × MEOP (7.0 MPa).
2. Water-fill with the motor vertical, AFT END UP: item 17's 1/8-27 NPT port (view B) is the motor's only port (item 6 has none) and its cup floor the cavity's highest point. Items 2 and 17 make one flat ceiling at z 38, so tilting would drive bubbles out to the bore: fill from the bottom instead (SET-UP schematic). Pass a ϕ4 × 800 PE or nylon dip tube, end cut at 45°, down the port and the ϕ6 passage to rest on item 6's inboard face; fill slowly so the water rises level. Withdraw it, top up until the port overflows, fit the rig tee (bleed valve on top, pump line on the side branch), pump slowly with the bleed open until only water comes out, then close it. Trapped air is stored energy.
3. Set the pump's relief valve to 11 MPa before pressurising. Stand behind a barrier while the motor is under pressure.
4. Pressurise in steps 3.5 → 7.0 → 10.5 MPa, holding 300 s at each. Accept a hold if the pressure decays ≤ 2 % over it and there is no visible leak or weep.
5. Measure the case OD at the marked stations (S1 … S5, SRM100-101 record R9) before and after. Accept a permanent set ≤ 0.02 mm at every station.
6. Re-gauge every bolt hole against its baseline (SRM100-101 record R7). Retire the end if any hole's axial length has grown by more than 0.05 mm, or its ligament has shrunk by more than 0.05 mm, from that hole's own baseline.
7. After the first proof, take the hardness / conductivity baseline on each case end (SRM100-101 record R8): HRB or Webster B, and eddy-current %IACS, at θ 50°, 130°, 230°, 310° (between the holes); aft end z 5–25 and z 100, fwd end z 665.8–685.8 and z 590.8. 6061-T6 reference: about 95 HB (about 60 HRB), Webster B ≥ 15, 40–46 %IACS. Later readings on the same instrument and scale; after a firing, retire the end if > 46 %IACS or hardness more than 10 % below this baseline (SRM100-101 R10).
8. Re-proof after any repair, any retired-and-replaced part, or a temperature label reading ≥ 232 °C.
9. Record against the serials of the case / closure pair and of item 17: the step pressures and hold decays; the OD before and after, the hole gauging and the hardness baseline on the SRM100-101 records R9, R7 and R8.
10. Rig tee into item 17's 1/8-27 NPT port (ASME B1.20.1): Loctite 567 or equal on the male thread, first 2 threads clear; no PTFE tape. By the test operator, not defined in the design data: gauge range and accuracy, the isolation valve that holds the pump off during the holds, valves, fittings and hoses, water quality and temperature, drain and dry-out.
11. After the test: drain through item 17's port, motor aft end down (or siphon out through the dip tube), then strip (PROOF BUILD note 10) and dry. Item 2's bore and gland must be dry before the graphite insert goes in. The flight build takes new O-rings and screws (items 13–16); the full replace / re-use list for every firing is on SRM100-000 sheet 2.

PROOF SET-UP (SCHEMATIC, NOT TO SCALE)

TEST ORIENTATION: AFT END UP



Gauge, tee, bleed and isolation valves, fittings and hoses: by the test operator (PROCEDURE note 10).

FITS AND SEALS (ref, from the detail drawings)

JOINT	HOLE / BORE (item)	SHAFT / LAND (item)	FIT (ref)
Case bore / closure lands	seal zones ϕA +0.030/0 (1)	ϕA − 0.06 0/−0.02 (2, 6)	clearance 0.060–0.110
Case O-ring glands, 4	groove ϕA − 5.67 0/−0.04 × 4.80 +0.08/0 (2, 6)	AS568-237 (13)	squeeze 19.7 %
Holder bore / dummy OD	ϕ63.58 +0.025/0 (2)	ϕ63.50 0/−0.05 (17)	clearance 0.080–0.155
Insert gland	groove ϕ69.14 +0.05/0 × 4.80 +0.08/0 (2)	AS568-230 (14)	squeeze 20.1 %
Seat depth / dummy length	seat 83.21 +0.05/0 (2)	83.21 0/−0.05 (17)	0–0.10 below face
Screw holes, 2 × 18	ϕ6.40 +0.03/0 (1)	M6 tapped (2, 6)	match-drilled pair

A = the smallest reading in each seal zone of item 1, 92.000–92.070, zone A +0.030/0 (so ≤ 92.10), per zone: SRM100-101 R4. Items 2 and 6 are cut to their own end's A; at A = 92.000: land ϕ91.94, groove ϕ86.33.

PROOF SCHEDULE (medium: water)

STEP	PRESSURE	HOLD	ACCEPT THE HOLD IF
1	3.5 MPa	300 s	decay ≤ 2 % over the hold, no visible leak or weep
2	7.0 MPa = MEOP	300 s	decay ≤ 2 % over the hold, no visible leak or weep
3	10.5 MPa = 1.5 × MEOP	300 s	decay ≤ 2 % over the hold, no visible leak or weep

Relief valve set 11 MPa before pressurising. Proof 10.5 MPa = 1.5 × MEOP 7.0 MPa.

AFTER THE TEST

CHECK	ACCEPT / ACTION
Case OD at stations S1 … S5 (SRM100-101 R9), before / after	accept a permanent set ≤ 0.02 mm at every station
Every bolt hole (2 × 18), against its baseline (SRM100-101 R7)	retire the end if any hole's axial length has grown, or its ligament has shrunk, by more than 0.05 mm from that hole's own baseline
Case ends, after the first proof	hardness / conductivity baseline, SRM100-101 R8 (PROCEDURE note 7)

Mass 6089.8 g, water not included: sum of qty × unrounded mass each. Items 1, 2, 6, 13–16 from out/mass_budget.json; item 17 from its STEP, adaptive BRepGProp volume × 8.00 g/cm³. Tooling (hole plugs, push sleeve, puller bridge, clamp collar) is not listed.

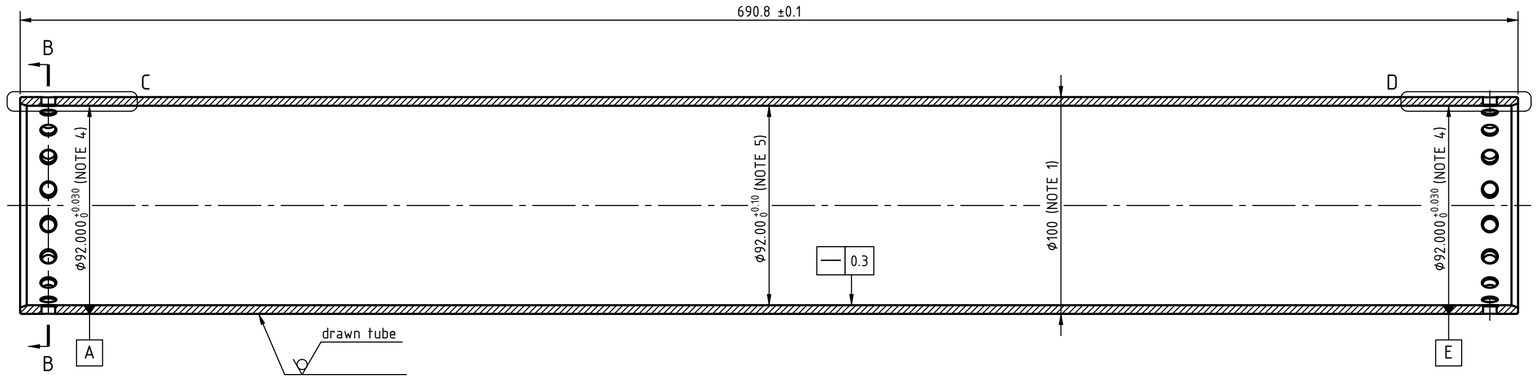
ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL	MASS EACH g
17	1	SRM100-120	Proof dummy (replaces item 3)	304 stainless, ASTM A479	1414.1
16	18	ISO 7380-1 M6×16-010.9	Button-head screw, fwd	steel 010.9	4.7
15	18	ISO 7380-1 M6×10-010.9	Button-head screw, aft	steel 010.9	3.4
14	1	AS568-230	O-ring, insert seal	FKM 90A (Parker V0709-90 or equal)	3.8
13	4	AS568-237	O-ring, case seal	FKM 90A (Parker V0709-90 or equal)	5.1
6	1	SRM100-103	Forward closure	6061-T651	770.5
2	1	SRM100-102	Nozzle holder (aft closure)	304 stainless	1501.2
1	1	SRM100-101	Motor case	6061-T6 drawn tube	2234.4

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
See parts list	6089.8 g	ISO 2768-mK	–	–

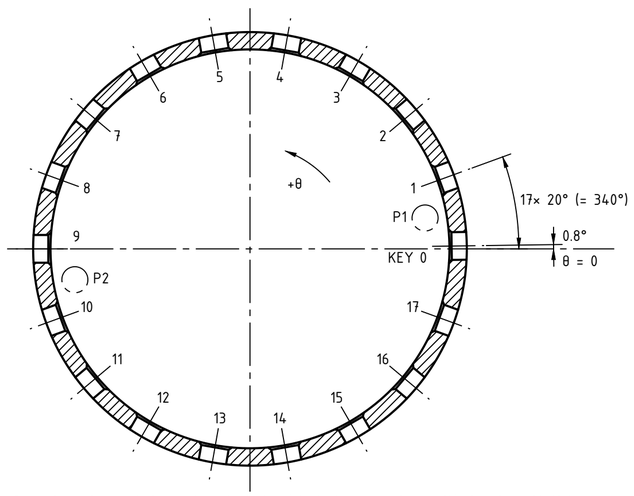
DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
ASSEMBLY DRAWING	NOT CHECKED – FOR REVIEW	1:2		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	HYDROSTATIC PROOF ASSEMBLY Flight case and closures, proof dummy SRM100-120 in the insert gland

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-001	A	1 / 1

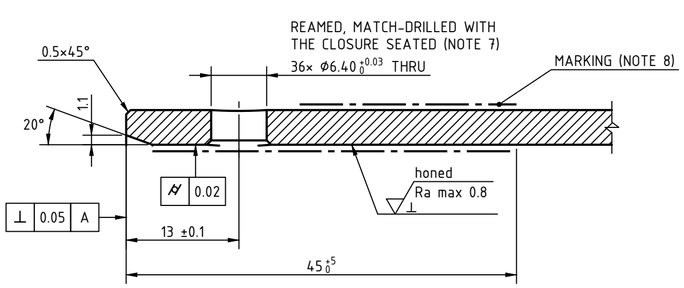


SECTION B-B (1:1)



- Seen from forward: θ counter-clockwise (clockwise seen from aft), θ = 0 = +X. Hole 0 (KEY) at 0.8°, holes 1 … 17 at 20° steps. Forward row (z 677.8, 13 ±0.1 from the fwd end face): the same pattern and numbering, its hole 0 in line with the aft hole 0 within ±2°.
- θ = 0 is set at match-drilling (op 120) from the closure's puller hole P1 at θ 10° (P2 at 190°; shown in phantom at the nozzle holder's R41, on its aft face; forward closure: R37.5, on its fwd face): hole 0 goes 9.2° clockwise of P1 seen from forward (9.2° counter-clockwise seen from aft). From then on hole 0, the clocking line and the AFT / FWD marking (note 8) carry θ = 0 on the case.

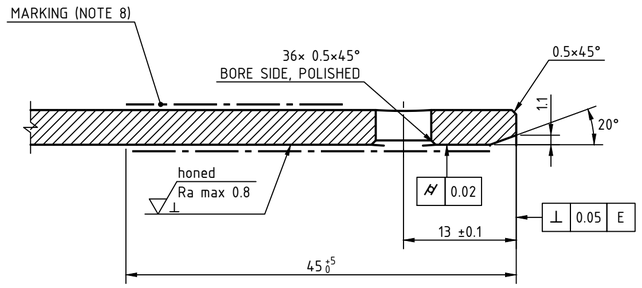
DETAIL C (2:1)



BOLT HOLES, EACH ROW

HOLE	θ ±0.1°	HOLE	θ ±0.1°
0 KEY	0.8°	9	180°
1	20°	10	200°
2	40°	11	220°
3	60°	12	240°
4	80°	13	260°
5	100°	14	280°
6	120°	15	300°
7	140°	16	320°
8	160°	17	340°

DETAIL D (2:1)



NOTES

1. Material: 6061-T6 seamless drawn tube to ASTM B210M, ϕ100 × 4.0 wall, mill certificate with heat / lot number (R1). OD and wall not machined: ϕ100 to the B210M tolerance, ISO 2768-mK does not apply to it. No heat treatment, welding, straightening or stamping.
2. Finish: none. The case is left bare (no anodising, paint or coating). Clean, oil-free, no marking ink or tape residue left on it.
3. Tube acceptance (op 40, R2, R3): wall ≥ 3.80 everywhere and bore ϕ ≤ 92.10 everywhere (two-point), the honed seal zones included (note 4 keeps them there). Reject the tube otherwise.
4. Seal zones (thick chain lines, details C, D; drawn at A = 92.000): hone each zone to the smallest diameter that cleans it up, A, and hold it to A +0.030/0, Ra 0.4 (Ra 0.8 max), circumferential lay. A = the smallest of the zone's six R4 readings, recorded per zone and marked next to it (note 8). Accept 92.000 ≤ A ≤ 92.070 (so no zone exceeds 92.10), else reject the tube. Closures finished to the A of their zone: plug land A − 0.06 (0/−0.02), O-ring groove bottom A − 5.67 (0/−0.04), i.e. 91.94 / 86.33 at A = 92.000 (SRM100-102, -103). Hone run-out blended into the as-drawn bore, no step. Datum A / E: axis of the aft / fwd honed zone.
5. Bore between the seal zones as drawn: ϕ92.00 … 92.10 (the clearance parts are sized to it), surface lines straight within 0.3. A GO plug ϕ91.70 × 150 and liner SRM100-110 must pass the full length by hand. A bore under 92.00 is honed to 92.00 … 92.10 full length.
6. Length 690.8 ±0.1 closes the axial stack (SRM100-000): 2 closure plugs, 2 insulators, 4 grains ±0.1, 5 spacers ±0.02 and the take-up ring, 3 installed. Stack ±0.5 + case ±0.1 gives a take-up gap of 2.4 … 3.6, i.e. 50 … 25 % compression of the 4.8 silicone sponge (limit 25 … 50 %). The gap is measured at assembly.
7. Bolt holes ONLY at assembly (ops 110 … 150), the mating closure seated. Each case end and its closure are a serialised pair: never interchange closures or turn the case end for end.
8. Marking (op 160), laser or dot-peen, 0.05 deep max, 3 mm characters: on the OD next to hole 0 at each end, z 20 … 45 from the end face, centred on θ = 0, reading round the tube: 'SRM100-101 S/N …' over 'AFT A 92.xxx' (aft end) or 'FWD A 92.xxx' (fwd end), A of that zone (note 4). Clocking line (op 140): scribed 0.05 deep max at θ = 0, z 0 … 8, continued across the closure flange. Temperature labels, fitted new at assembly for every firing: irreversible 177 / 204 / 232 / 260 °C strips on the OD at z 20 … 35 from the aft face, θ 90° and 270°, clear of the marking and head sweep (two more on the holder, SRM100-102); read and recorded before extraction (R10).
9. Re-use: the case is re-used after inspection (R10) after every firing. Proof test on SRM100-001 (R9) before first use and after any repair, any retired-and-replaced closure, or a temperature label reading ≥ 232 °C. The case record R1 … R10 stays with the case serial for its life, with the closure pair serials.
10. Edges ISO 13715 -0.3 / +0.3 unless stated.
11. USED ON: SRM100-000 motor, item 1 (1 off); SRM100-001 hydrostatic proof assembly, item 1.

OPERATIONS

OP	LOOSE TUBE (as supplied to assembly)	OP	AT ASSEMBLY SRM100-000, each end with its closure seated
10	Receive the tube with its mill certificate (note 1, R1). Cut and face both ends to 690.8 ±0.1; end faces square to the bore (⊥ 0.05 A / E, checked after op 30).	110	Take the AFT end (tape label) and the nozzle holder SRM100-102 finished to its A (FWD end: forward closure SRM100-103). Seat the closure in the bore, flange on the end face, O-rings not fitted. Clamp the pair.
20	Turn the 0.5×45° OD break and bore the 20°×1.1 radial O-ring lead-in at both ends (details C, D); lead-in blended into the bore, no step, Ra 0.8.	120	Clock hole 0 from the closure's puller hole P1 (θ 10°): 9.2° clockwise of P1 seen from forward, 9.2° counter-clockwise seen from aft (SECTION B-B). Drill ϕ5 (tap drill) through the case wall into the closure, depth stops per SRM100-102 / -103: hole 0 first, then holes 1 … 17.
30	Label the ends AFT and FWD (removable tape). Measure each seal zone (two-point). Hone the zone, 45 +5/0 from the end face, to the smallest diameter that cleans it up: Ra 0.4 (0.8 max), circumferential lay, cylindricity 0.02, run-out blended into the as-drawn bore. A = the smallest of its six R4 readings; accept 92.000 ≤ A ≤ 92.070, zone A +0.030/0 (note 4). Record A (R4) and write it on the tape label.	130	Case wall only, closure still in place: drill ϕ6.2 and ream ϕ6.40 +0.03/0, piloted in the ϕ5 hole; stop when the full diameter is through the case bore (the entry cone left in the closure, 0.5 deep max, lies in its first, uncounted thread). Tap the closure M6 through the case holes (per SRM100-102 / -103).
40	Inspect and record (R2 … R6): UT wall ≥ 3.80 and bore ϕ ≤ 92.10 (note 3); seal zones A +0.030/0; mid bore, straightness, GO plug and liner trial (note 5); length and ⊥.	140	Scribe the clocking line at θ = 0 across the case end and the closure flange; mark the closure per SRM100-102 note 8 / SRM100-103 note 11 (pair serial included); the case end is marked at op 160 (note 8). Remove the closure.
50	Clean (IPA), cap the ends, tag with the case serial and both A values. The tube leaves this stage with NO bolt holes and NO permanent marking; the closures are finished to the A of their end (SRM100-102, -103).	150	Break every hole on the bore side 0.5×45° and polish (the O-rings are dragged over it); deburr the OD edges. Remove all chips; re-inspect the seal zone. Repeat ops 110 … 150 at the FWD end, the forward closure turned so that its P1 lies 9.2° counter-clockwise (seen from forward) of the aft hole 0 line, transferred along the tube, ±2°.
		160	Remove the tape labels and mark the case (note 8). Record the hole baseline (R7). Proof test on SRM100-001 (R9), then take the hardness / conductivity baseline (R8).

CASE RECORD (note 9)

REC	CHECK	WHEN	WHERE / HOW	ACCEPT
R1	Mill certificate	op 10	ASTM B210M, 6061-T6, heat / lot number	per B210M
R2	Wall, ultrasonic map	op 40	8 azimuths every 50 along the full length; all round each bolt row, z 5 … 21 from each end face	wall ≥ 3.80 everywhere
R3	Bore, two-point	op 30 (pre-hone), op 40	every 50 along the length, 2 directions 90° apart	≤ 92.10 everywhere, honed zones included (R4); mid bore note 5
R4	Seal zones AFT, FWD: A and diameters	op 30, op 40	z 10, 25, 40 from the end face × 2 directions 90° apart, to 0.001: A = the smallest of these 6 readings, per zone; Ra; A also marked on the tube (note 8)	92.000 ≤ A ≤ 92.070; zone A +0.030/0 (so ≤ 92.10); Ra ≤ 0.8
R5	Mid bore	op 40	surface-line straightness; GO plug ϕ91.70 × 150 over the full length; liner SRM100-110 slide trial	note 5
R6	Length, end faces	op 40	length at 4 azimuths; each end face to its datum A / E	690.8 ±0.1; ⊥ 0.05
R7	Hole baseline, 36 holes	op 160, before the first proof; after every proof	per hole: axial and circumferential ϕ, ligament end face to hole edge (nominal 13 − 3.2 = 9.80; a new in-tolerance case can start at 12.9 − 3.215 = 9.685, so no absolute limit)	retire the end if any hole's axial length has grown, or its ligament has shrunk, by more than 0.05 from that hole's own baseline
R8	Hardness and conductivity baseline	op 160, after the first proof	HRB or Webster B, and eddy-current %IACS, at θ 50°, 130°, 230°, 310° (between the holes): aft end z 5 … 25 and z 100; fwd end z 665.8 … 685.8 and z 590.8	record, same instrument and scale for life (6061-T6: about 95 HB (about 60 HRB), Webster B ≥ 15, 40 … 46 %IACS)
R9	Proof OD, SRM100-001	before and after every proof	stations S1 … S5 at z 60, 172.7, 345.4, 518.1, 630.8 from the aft face, on removable tape; diameters through θ 45° and 135°, micrometer 0.001, tube temperature at the two readings within 2 °C	permanent set ≤ 0.02 at every station
R10	After every firing, before re-use	each firing	temperature labels (note 8), read and recorded before extraction; hardness and %IACS on the aft end at the R8 positions; holes as R7; seal zones visual (no scratch across the lay)	label ≥ 232 °C: re-proof or retire that end; > 46 %IACS, or hardness more than 10 % below its R8 baseline on the same instrument and scale: retire

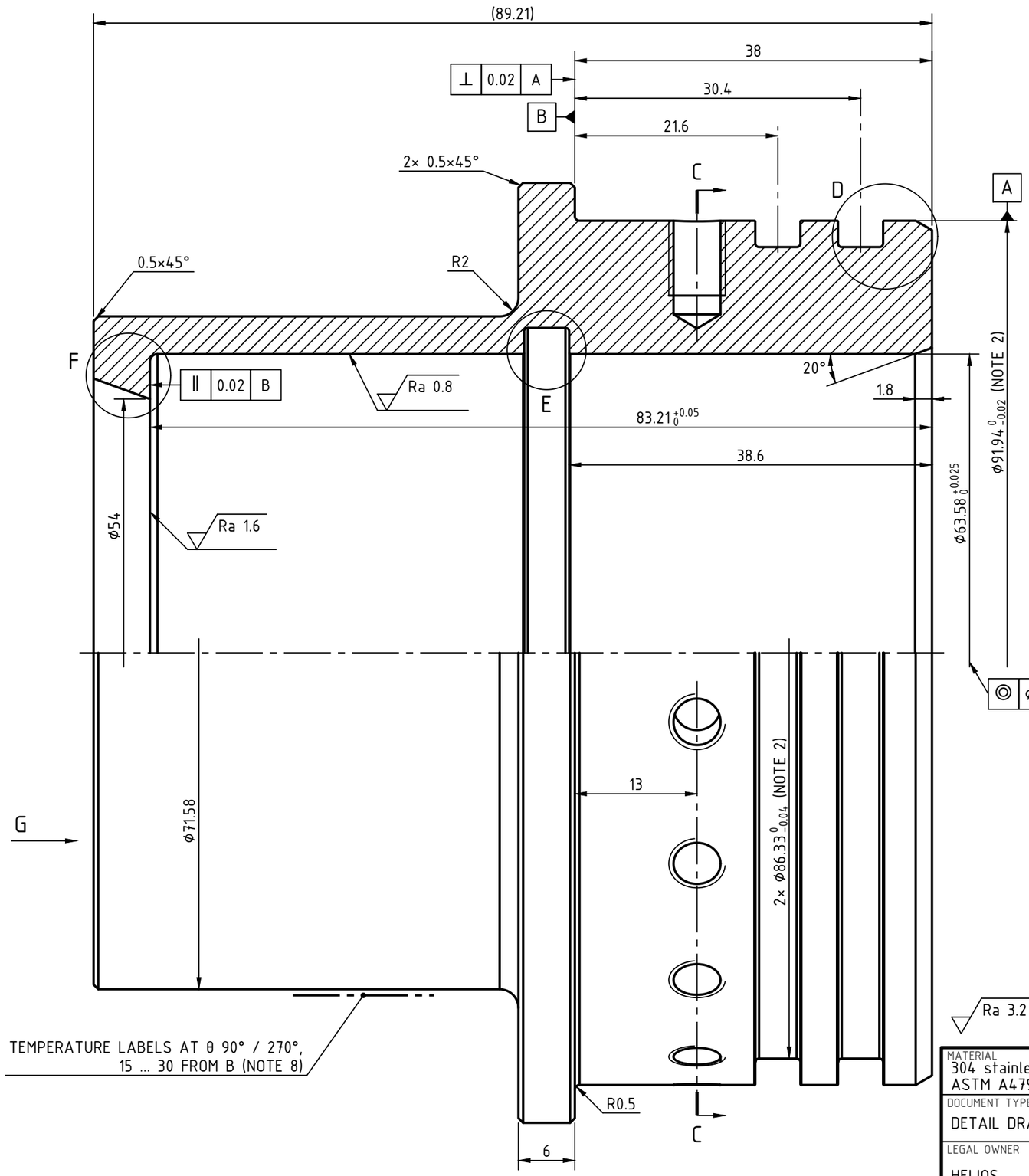
Ra 3.2 (✓)

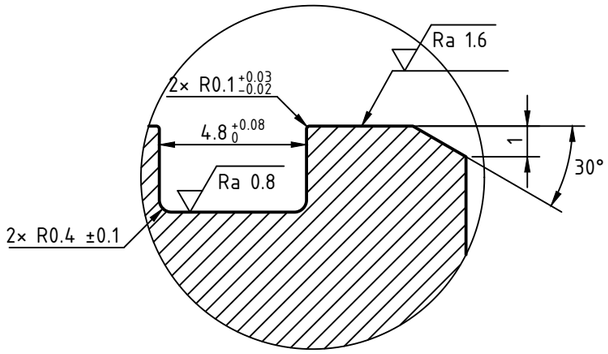
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
6061-T6 drawn tube, ASTM B210M	2234.4 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	1:2		mm

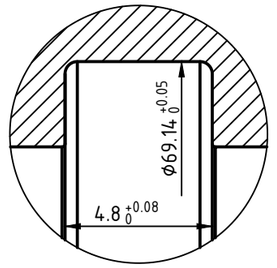
LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	MOTOR CASE Case tube ϕ100 × 4 × 690.8, bare 6061-T6, 2 × 18 radial bolt holes

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-101	A	1 / 1

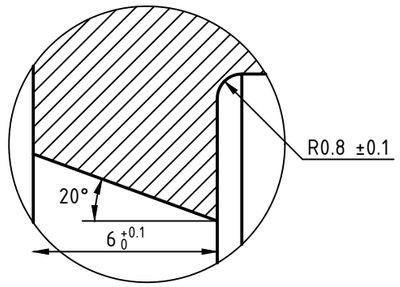




DETAIL D (5:1)
BOTH CASE GROOVES ALIKE, WALLS Ra 1.6



DETAIL E (5:1)
R0.4 AND R0.1 AS DETAIL D



DETAIL F (5:1)

SEAL DIAMETERS FROM THE MEASURED CASE BORE A (NOTE 2)

FEATURE	CUT TO	A = 92.000 (DRAWN)	A = 92.070 (MAX)
Land (datum A)	A − 0.06 0/−0.02	ϕ91.94 0/−0.02	ϕ92.01 0/−0.02
Case grooves (2×)	A − 5.67 0/−0.04	ϕ86.33 0/−0.04	ϕ86.40 0/−0.04
Case bore A	recorded: _____	92.000	92.070

A = the smallest reading in the case's aft seal zone, honed and recorded on SRM100-101 (op 30, R4): 92.000 ≤ A ≤ 92.070; the zone is held to A +0.030/0, ≤ 92.10.

Upper half: section in the plane of hole 0 (θ 0.8°). SECTION C-C, VIEW G and the NOTES (for both sheets) are on sheet 2.

Ra 3.2 (✓)

USED ON: SRM100-000 item 2; SRM100-001 item 2

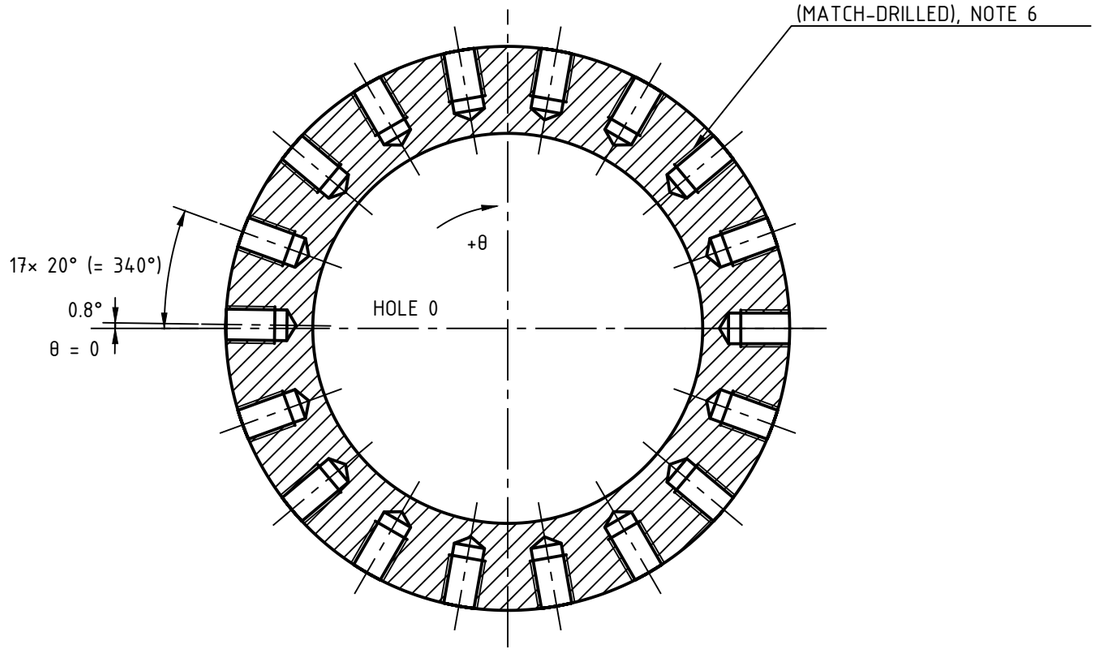
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
304 stainless (1.4301) ASTM A479 annealed bar	1501.2 g	ISO 2768-mK	Ra 3.2	ISO 13715 −0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	2:1		mm

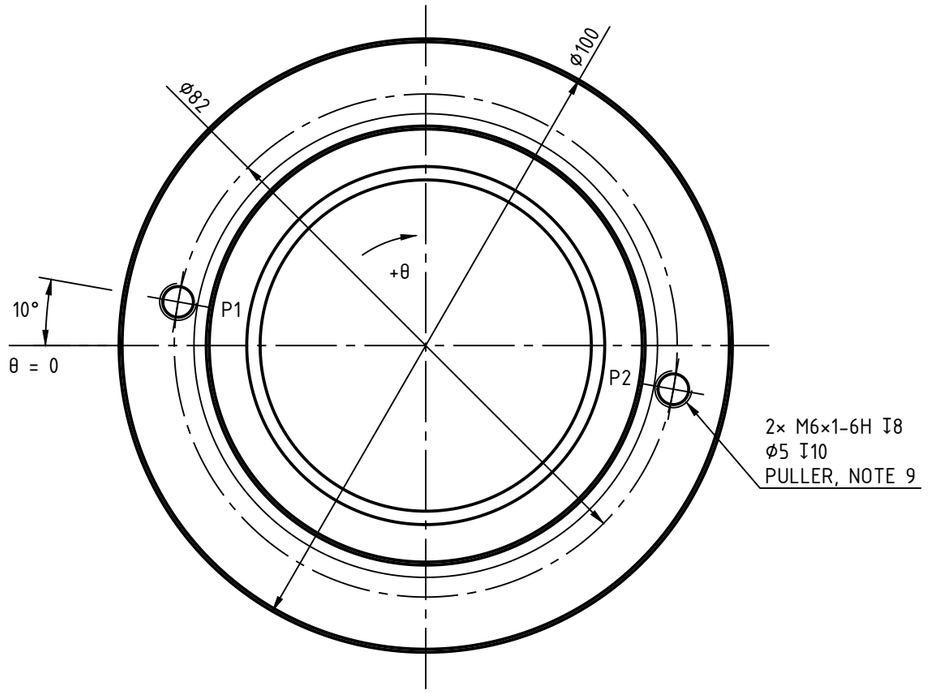
LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	NOZZLE HOLDER Aft closure – 18× M6 radial – insert bore ϕ63.58

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-102	A	1 / 2

SECTION C–C



VIEW G



NOTES (SHEETS 1 AND 2)

1. Material: stainless steel 304 (1.4301, X5CrNi18-10), ASTM A479 annealed round bar ⌀105 min × 95 min, EN 10204 3.1 certificate. Substitutes (303 included) only with engineering approval.
2. The holder is cut to its own case: land ⌀(A − 0.06) 0/−0.02, case grooves ⌀(A − 5.67) 0/−0.04 (table, sheet 1), A being the honed size of the mating case's aft seal zone, recorded on SRM100-101 (note 4, R4): the smallest reading in the zone, 92.000 ≤ A ≤ 92.070, the zone held to A +0.030/0 (never over 92.10). ⌀91.94 and ⌀86.33 are drawn for A = 92.000. Record A, the cut sizes and the pair serial.
3. Finish-turn the land, both case grooves, the insert bore, the seat face and face B in one setup.
4. Seat root R0.8 ±0.1: smooth blend, no undercut, no step (the bending root of the lip). Glands: bottoms Ra 0.8, walls Ra 1.6, no tool marks across a groove; groove edges R0.1 (0.08 – 0.13).
5. Glands for 2× AS568-237 (case seal, on the land) and 1× AS568-230 (insert seal, in the bore), SRM100-000 items 13 and 14: FKM 90A (Parker V0709-90 or equal), new for every firing. Lubricant: Parker Super O-Lube or DC-55, thin film.
6. 18× M6×1-6H radial holes, made at assembly per SRM100-101 ops 110 – 150, the holder seated and clamped in its case, hole 0 first: tap drill ⌀5 through the case holes, depth stops set from the case OD (the land is covered): full ⌀5 to 14.0 (9.97 from the land), full thread to 12.0 (7.97 from the land); drill point ≥ 2 from the insert bore (2.71 nominal). Bottoming tap through the reamed case holes; M6-6H GO / NO-GO gauge. The SRM100-101 op 130 ream entry cone is permitted (0.5 deep max, ⌀6.40 at the land); full thread is counted below it. Case and holder are then a serialised pair.
7. Clocking: θ is counter-clockwise seen from forward, clockwise seen from aft (SECTION C–C, VIEW G). Until match-drilling θ = 0 is carried by puller P1 (θ 10°, ⌀82): hole 0 (key, θ 0.8°) is 9.2° clockwise of P1 seen from forward, 9.2° counter-clockwise seen from aft. The clocking line at θ = 0 is then scribed across the case end and the flange OD (SRM100-101 ops 120, 140).
8. Mark 'SRM100-102 AFT', the part serial, the pair serial and A (note 2) on the flange OD, centred on hole 0 (θ 0.8°), a 2 mm gap left in the text where the clocking line θ = 0 crosses: 2 mm characters, laser or dot-peen, 0.05 deep max. No marks on the aft body OD ⌀71.58: it carries the temperature labels (sheet 1), irreversible 177 / 204 / 232 / 260 °C strips fitted new at assembly for every firing, at θ 90° and 270°, 15 … 30 from B, clear of the cap tape (SRM100-106 note 6: at θ 0° / 180°, ends ≤ 15 from the aft face).
9. Puller threads P1 / P2 (VIEW G): 2× M6×1-6H ↧8, tap drill ⌀5 ↧10, in the flange aft face on ⌀82 at θ 10° (P1) and 190° (P2); machined in the shop, before match-drilling. For the puller bridge and two draw screws only (tooling). Never lever or knock the holder out of the case.
10. Aft case screws: 18× SRM100-000 item 15, ISO 7380-1 M6×10-010.9, new for every firing. Nickel anti-seize on the threads only, head face dry; 3.0 – 3.5 N·m in the criss-cross sequence of SRM100-000 sheet 2 (3 passes). 6 N·m with anti-seize yields the annealed 304 threads. Never fit an M6×16 here: it bottoms with the head proud.
11. Insert removal: support the holder on its forward face; the push sleeve (tooling, 52.4 ID × 53.8 OD) bears on the insert's aft face through the lip bore. Never push on the lip.
12. Proof test per SRM100-001 with the proof dummy SRM100-120 in place of the insert.
13. Inspect 100 %: land, case grooves and insert groove (⌀ and width), insert bore, seat depth, lip thickness 6 +0.1/0, the geometric tolerances; the glands and the seat root visually at 10×. Record the values with the serial.
14. Deburr and clean: every edge an O-ring passes over is burr-free and smooth (nose chamfer, land, case groove edges, bore lead-in, insert bore, insert groove edges). Tapped-hole entries on the land flush, no raised burr (the land slides in the honed bore). After machining and again after match-drilling: blow out, wash and degrease (IPA); no chips in the glands, threads or bore.
15. After every firing: read and record the temperature labels before extraction; a reading ≥ 232 °C: retire or re-proof (SRM100-001) the aft end. Replace and re-use per SRM100-000 sheet 1 note 13; item 2 is re-used after inspection: clean and check the glands and seat face (no erosion, pitting, scoring), seat root and lip at 10× (no crack), lip thickness and seat depth, all tapped holes with the M6-6H GO / NO-GO gauge. Any finding: quarantine, engineering decides.
16. Surface texture indications per ISO 21920-1 (Ra, ISO 21920-2).

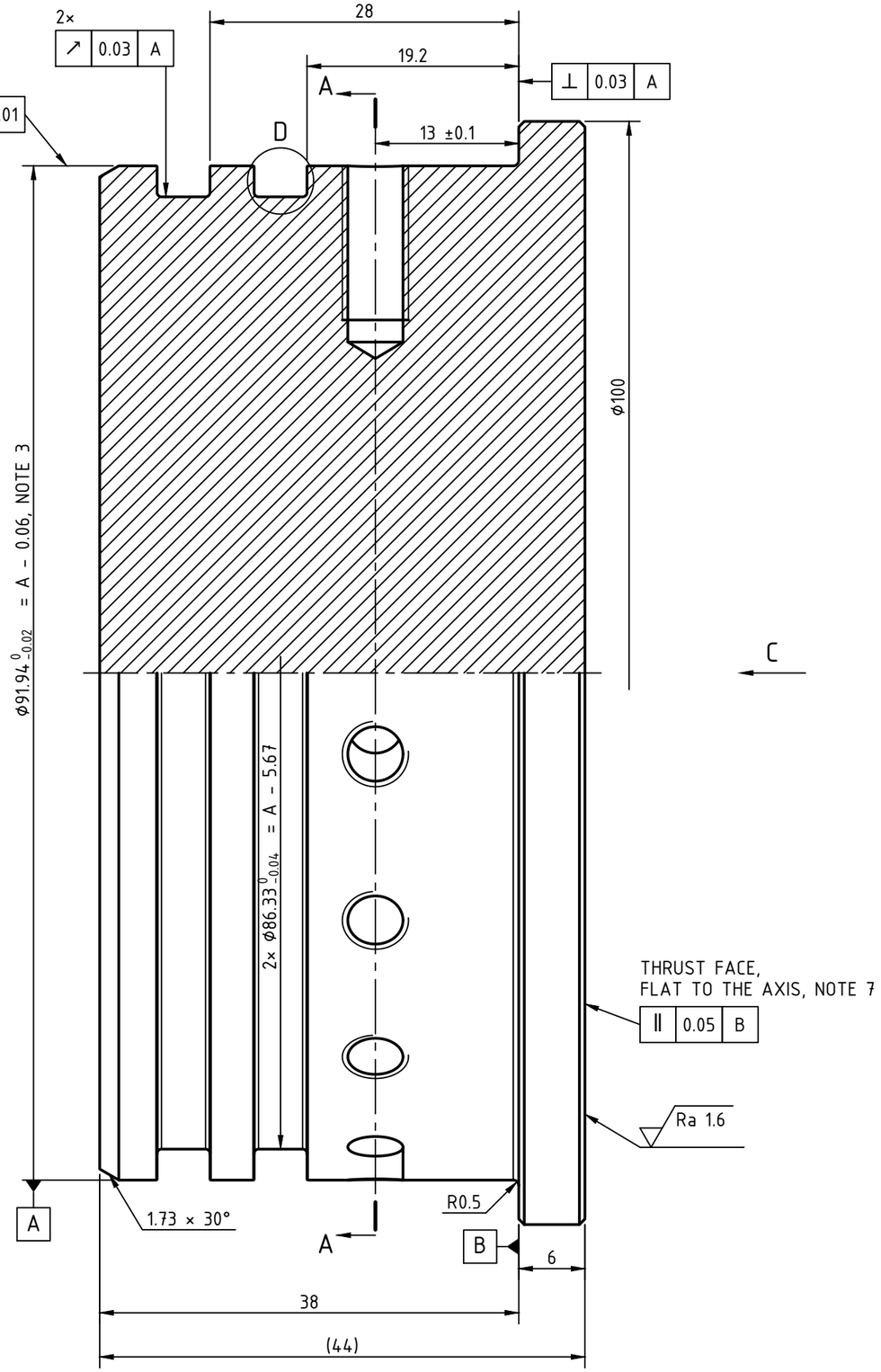
USED ON: SRM100-000 item 2; SRM100-001 item 2

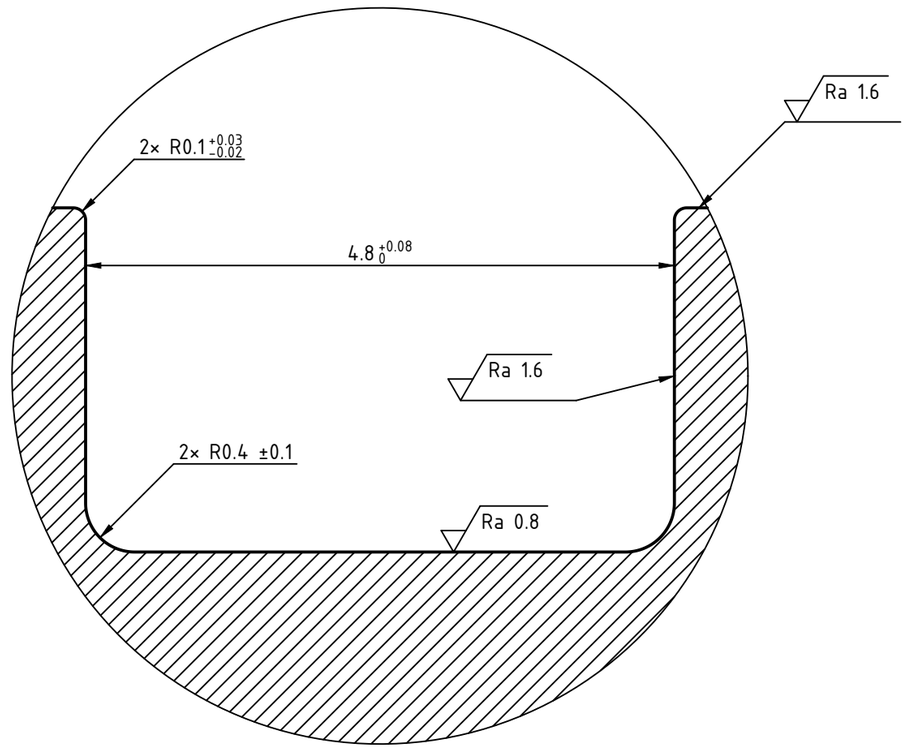
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
304 stainless (1.4301) ASTM A479 annealed bar	1501.2 g	ISO 2768-mK	Ra 3.2	ISO 13715 −0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	NOZZLE HOLDER Aft closure – 18× M6 radial – insert bore ⌀63.58

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-102	A	2 / 2





D (20:1)
BOTH GROOVES

Ra 3.2 (✓)

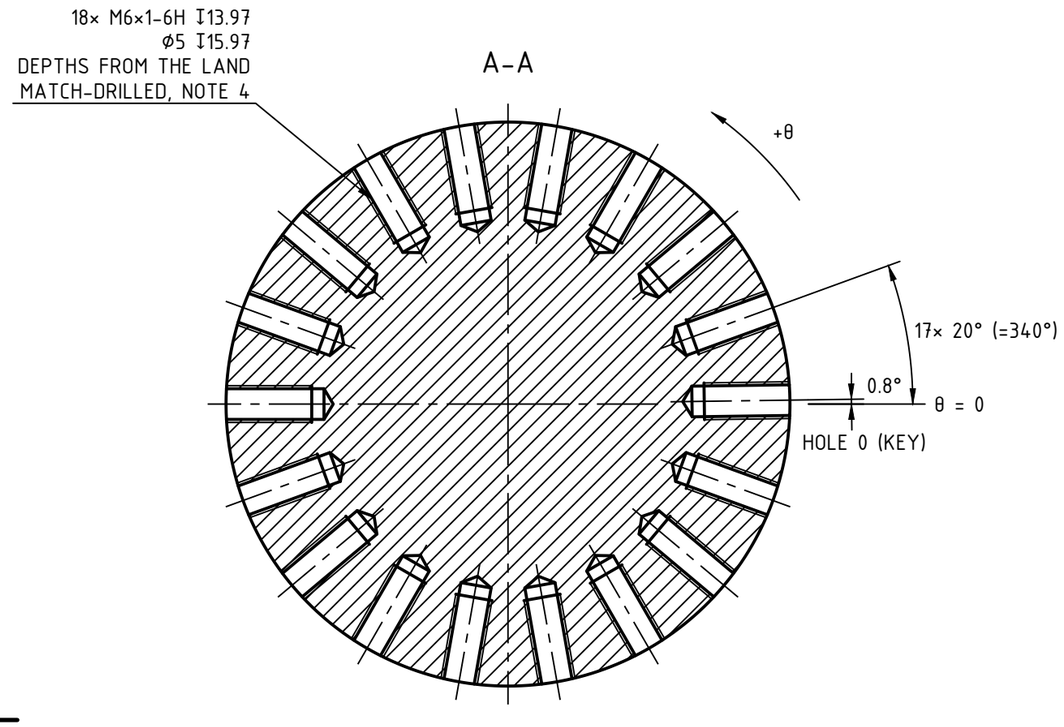
NOTES, MATCH-DRILLING AND SECTION A-A: SHEET 2
USED ON: SRM100-000 item 6 (1 off); SRM100-001 item 6

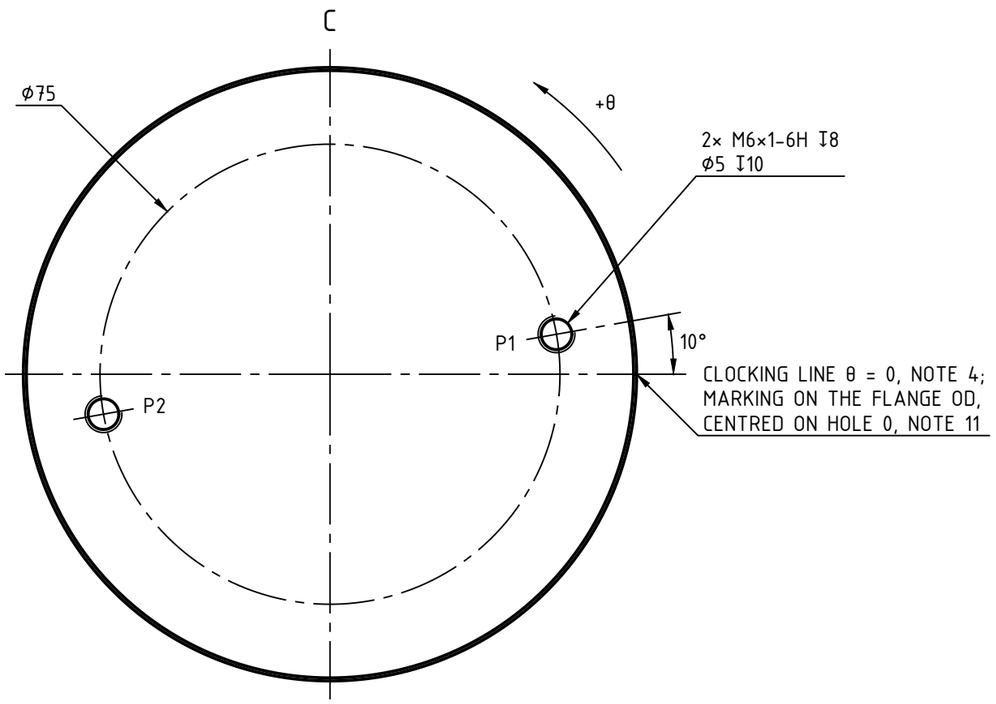
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
6061-T651 bar, ASTM B211/B221 or equal, stock ⌀105 × 50 min	770.5 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	2:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	FORWARD CLOSURE Half section (solid, flat forward face), seal groove D

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-103	A	1 / 2





SECTION A–A AND VIEW C ARE SEEN FROM FORWARD: +θ COUNTER-CLOCKWISE, θ = 0 ON +X.
P1 AT θ = 10°, P2 AT 190°; HOLE 0 (θ = 0.8°) IS 9.2° CLOCKWISE OF P1.

NOTES

1. Material: aluminium alloy 6061-T651 round bar to ASTM B211 or ASTM B221 or equal (AMS 4117; B221 extruded bar may come as 6061-T6511), stock ⌀105 × 50 min, grain along the axis. Mill certificate with the heat / lot number required.
2. Bare metal: no anodizing, conversion coat or paint. Texture: land and groove walls Ra 1.6, groove bottoms Ra 0.8, thrust face Ra 1.6, elsewhere Ra 3.2; circumferential lay on the land and grooves. Finish the land, both grooves and face B in one setup.
3. The plug is cut to its own case: land ⌀(A − 0.06) 0/−0.02, both groove bottoms ⌀(A − 5.67) 0/−0.04, where A is the honed size of the mating case's forward seal zone as recorded on SRM100-101 (note 4, R4): the smallest reading in the zone, 92.000 ≤ A ≤ 92.070, the zone held to A +0.030/0 (never over 92.10). ⌀91.94 and ⌀86.33 are drawn for A = 92.000. Record the A used with the serial.
4. Radial holes: made ONLY at assembly by match-drilling, per SRM100-101 ops 110 … 150, this closure seated and clamped in its case (face B on the case forward end face, O-rings out), hole 0 first. Until then θ = 0 is carried by puller P1 (θ = 10°): hole 0 (θ = 0.8°) is 9.2° clockwise of P1 seen from forward (view C). Depth stops from the case OD (the land is covered): full ⌀5 to 20.0 (15.97 from the land), full thread to 18.0 (13.97); bottoming tap through the reamed case holes. The ream entry cone of SRM100-101 op 130 is permitted in each hole (0.5 deep max, ⌀6.40 at the land); full thread is counted below it. Scribe the clocking line θ = 0 across the case end and the flange OD, 0.05 deep max. Case and closure are then a serialised pair: never interchange.
5. The 18 radial tap drills end at r 28.50 in solid metal; the closest pair (holes 0 and 1, 19.2° apart) leave 4.51 between their points, 2.0 min.
6. Undimensioned chamfers 0.5 × 45° (flange edges); groove edges per detail D. Every edge an O-ring passes over burr-free and smooth; tapped-hole entries on the land flush, no raised burr. After machining and again after match-drilling: blow out, wash and degrease (IPA); no chips in the grooves or threads.
7. No port and no pocket: item 6 is solid on the axis, 44 thick, and its forward face is one flat thrust face, broken only by the 2 M6 puller holes (note 10). Do not drill, tap or pocket it beyond this drawing. The motor is fired without a chamber transducer; a transducer needs a new, ported closure (new drawing) and a re-proof.
8. Item 7 (SRM100-107, EPDM full disc ⌀91.8 × 3) is bonded to the aft face only after the proof test (SRM100-000 sheet 3): leave that face as machined, unmarked. IPA degrease, abrade (P180 / Scotch-Brite), IPA wipe, silicone primer SS4004P, RTV106 or equal over the whole face, bondline 0.1 … 0.2, disc OD centred on the land within 0.1; cure 24 h.
9. O-rings 2 × AS568-237, FKM 90A (Parker V0709-90 or equal), item 13; Parker Super O-Lube or DC-55, thin film. Screws 18 × ISO 7380-1 M6×16-010.9, item 16, Loctite 243, 4 … 6 N·m, criss-cross in 3 passes (SRM100-000 sheet 3).
10. Pullers P1 (θ = 10°) and P2 (190°) on ⌀75: 2 × M6×1-6H ↧8, ⌀5 ↧10, machined before match-drilling; for the puller bridge and two draw screws only (tooling). Radial screws out first; never lever or knock the closure out.
11. Mark 'SRM100-103 FWD', the part serial, the pair serial and A on the flange OD, centred on hole 0 (θ 0.8°), a 2 mm gap left in the text where the clocking line θ = 0 crosses: 2 mm characters, laser or dot-peen, 0.05 deep max. No marks on the thrust face, face B, land, grooves or aft face.
12. Inspect 100 %, record with the serial: land ⌀ (3 positions × 2 directions), groove bottom ⌀ and widths, the geometric tolerances, grooves visually at 10×; after match-drilling every radial thread with the M6-6H GO / NO-GO gauge, and its depth.
13. After every firing (SRM100-000 items): replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels; re-use items 1, 2, 6 after inspection (SRM100-101 R10 for the case), and item 3 while the throat is ≤ ⌀23.9 and uncracked. This closure (item 6): remove item 7 and the old RTV with a plastic scraper (no metal tools), clean, and repeat the note 12 checks: no erosion, heat damage or scoring on the land, grooves or aft face, threads in gauge. Any finding: quarantine, engineering decides.

Ra 3.2 (✓)

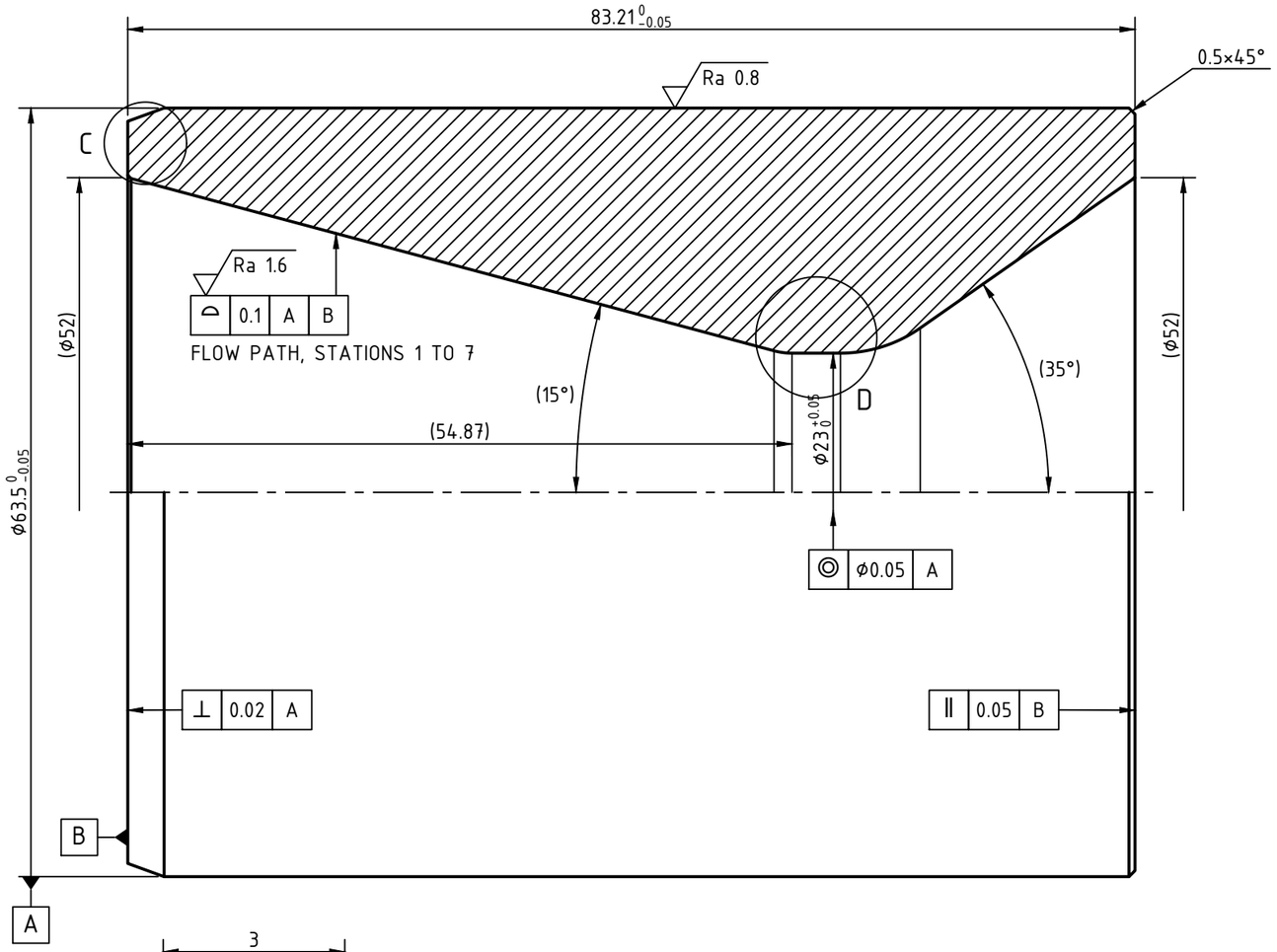
USED ON: SRM100-000 item 6 (1 off); SRM100-001 item 6

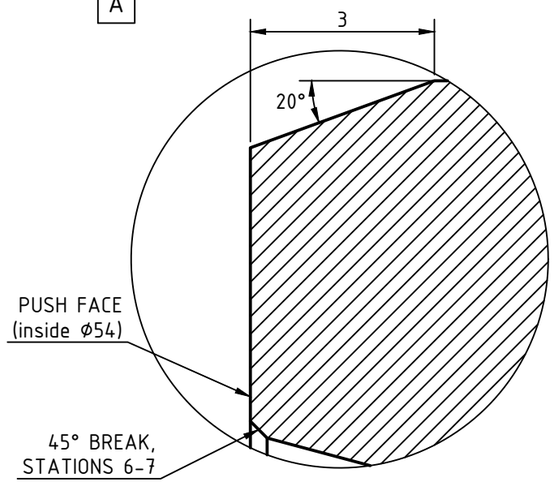
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
6061-T651 bar, ASTM B211/B221 or equal, stock ⌀105 × 50 min	770.5 g	ISO 2768-mK	Ra 3.2	ISO 13715 −0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

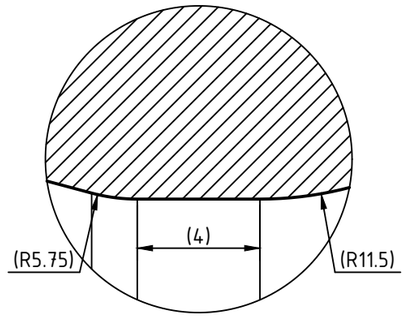
LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	FORWARD CLOSURE Section A–A (radial tapped holes), view C, notes

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-103	A	2 / 2





DETAIL C (10:1)



DETAIL D (5:1)

NOTES

1. Material: isostatically moulded fine-grain graphite, grade ISO-63, TTK-4, IG-43 or equal: density ≥ 1.78 g/cm³, flexural strength ≥ 50 MPa, grain ≤ 15 µm; not extruded or vibration-moulded. Certificate required. Blank ϕ68 × 88 min from one block. Mass 311.1 g at 1.80 g/cm³.
2. Machine dry. Never wet this part, never hydro-test it: the proof test uses the steel proof dummy SRM100-120 in its place. No tool marks across the throat land.
3. Edges: forward OD 0.5×45°; aft OD 20° × 3 lead-in; exit 45° break (stations 6-7). Entry edge (station 1) sharp, burr-free, ISO 13715 -0.05, to meet insulator SRM100-105 with no step. Throat blends tangent: do not break. Other edges per the title block.
4. Slip fit in holder SRM100-102 bore ϕ63.58 +0.025/0 (clearance 0.08 to 0.155): by hand, aft end first; never press or knock it. Seated on the lip (seat 83.21 +0.05/0) the forward face lies flush to 0.1 below the holder face. Seal: O-ring AS568-230, FKM 90A (Parker V0709-90 or equal), in the holder gland; grease Parker Super O-Lube or DC-55, thin film.
5. Inspect 100 %: OD and throat at three axial positions, length, contour to the table, coaxiality, perpendicularity, parallelism; 10× visual of the flow path and OD: no cracks, chips or pits. Record against the serial.
6. Laser-mark SRM100-104 and serial on the forward face, between ϕ56 and ϕ62, 2 mm characters, 0.05 deep max (no dot-peen: brittle graphite).
7. Clean with dry, oil-free compressed air only: no liquids, solvents or ultrasonic bath (porous). Handle with clean gloves; store bagged with desiccant.
8. Re-usable. After each firing measure the throat at three axial positions and enter it as THROAT_D (README). Re-use while the throat is ≤ ϕ23.9 and uncracked, with no chips on the flow path or the OD seal surface; otherwise scrap.
9. Used on SRM100-000 item 3. Not fitted on proof assembly SRM100-001 (proof dummy SRM100-120 in its place).

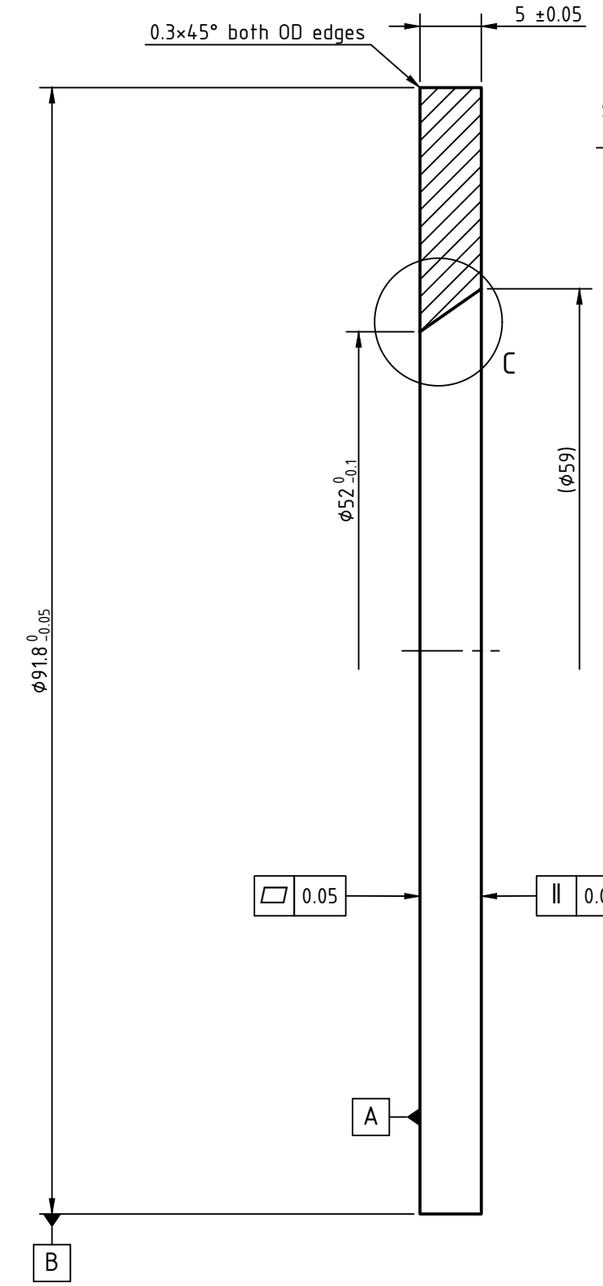
FLOW-PATH CONTOUR (THEORETICALLY EXACT)

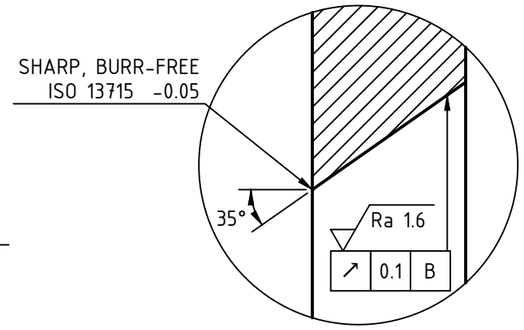
STATION	z FROM B	ϕ	NOTE
1	83.206	52.000	entry, forward face: sharp edge (note 3)
2	65.468	27.160	35° cone / R11.5 tangent
3	58.872	23.000	R11.5 / land tangent (throat area 415.5 mm²)
4	54.872	23.000	land / R5.75 tangent
5	53.384	23.392	R5.75 / 15° cone tangent
6	0.273	51.854	15° cone / 45° exit break
7	0.000	52.400	45° exit break / aft face B
E	0.000	52.000	exit, 15° cone meets B (construction; area ratio 5.11)
C1	58.872	46.000	centre of R11.5
C2	54.872	34.500	centre of R5.75

Table values are TEDs (profile frame); values in parentheses in the views are auxiliary.

Ra 3.2 (✓)

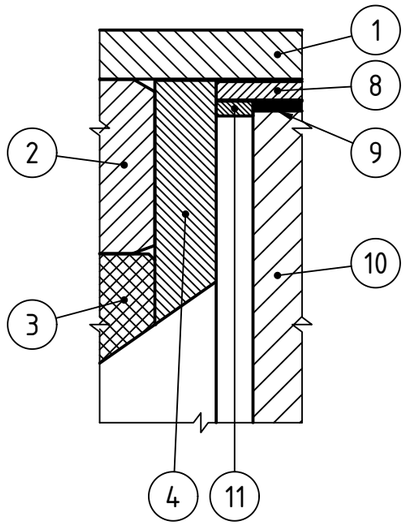
MATERIAL: Isomolded graphite, ISO-63 class	MASS: 311.1 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: Ra 3.2	EDGES: ISO 13715 -0.3 / +0.3
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 2:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: NOZZLE INSERT — Throat ϕ23 - 35° / 15° conical de Laval - graphite			
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-104	REV: A / SHEET: 1 / 1





DETAIL C (5:1)

INSTALLATION (REFERENCE) 2:1



NOTES

1. Material: glass-fabric reinforced phenolic laminate, NEMA LI 1 grade G-3, or compression-moulded glass-phenolic (MXB-360 / Durite class); density about 1.85 g/cm³. G-10 / G-11 are epoxy-glass grades and are not acceptable. Certificate of conformity.
2. Stock: sheet 6.35 (1/4 in) min, faced both sides to 5; laminations parallel to the faces. Turn OD and bore in one set-up on a mandrel located on datum A.
3. Machine dry with carbide or PCD tools, with dust extraction. No coolant, no oil. Blow clean with dry, oil-free air; deliver and store bagged, clean and dry.
4. The bore edge on the aft face is sharp, free of burrs and chips (detail C): it meets the graphite entry ϕ52 of insert SRM100-104 with no forward-facing step. ϕ52 is never exceeded.
5. Edges: OD edges 0.3×45° (not in the 3-D model): the part slides through the honed seal bore of case SRM100-101 at assembly. Other edges per the title block.
6. Loose fit in the aft honed zone of case SRM100-101, bore ϕ92.000 to 92.10 (A +0.030/0, A ≤ 92.070, SRM100-101 note 4): diametral clearance 0.20 to 0.35. Not bonded; located radially by the case bore. Aft face (A) down on the holder and graphite forward faces; clamped axially by the aft grain spacer on its forward face (grain stack, take-up ring). Liner SRM100-110 also rests on its forward face and carries no axial load. Grease the OD and edges: Parker Super O-Lube or DC-55, thin film.
7. Inspect 100 %: thickness at 4 places 90° apart, OD, bore ϕ52 at the aft face (optical comparator or taper gauge), flatness and parallelism on a surface plate, runout on a mandrel. Visual: no delamination, cracks or chipped edges. Record values.
8. Mark SRM100-105 and serial on the aft face between ϕ70 and ϕ86, indelible ink or laser, no engraving.
9. Single use: a new part for every firing, scrapped after it (charred bore; README, After every firing). Never re-use a fired part.
10. Used on SRM100-000 item 4 (aft closure bench subassembly). Item numbers in the installation view per SRM100-000.

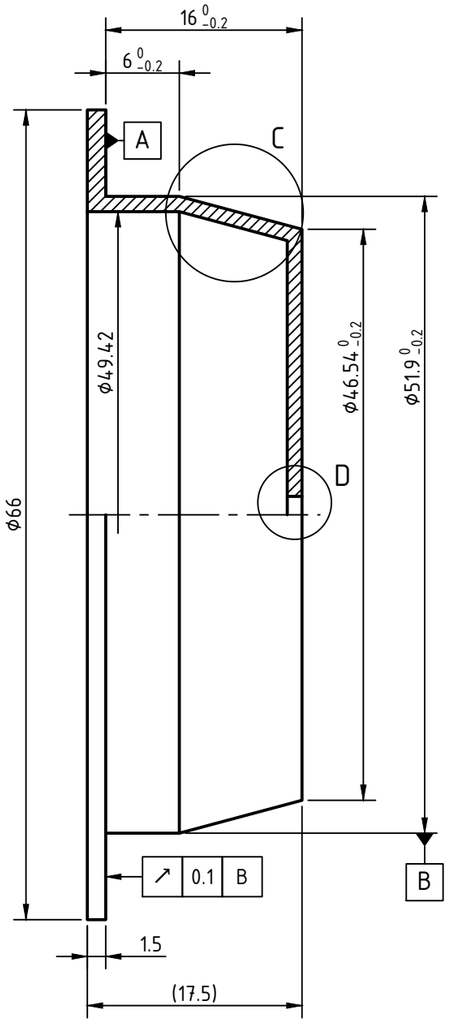
Ra 3.2 (✓)

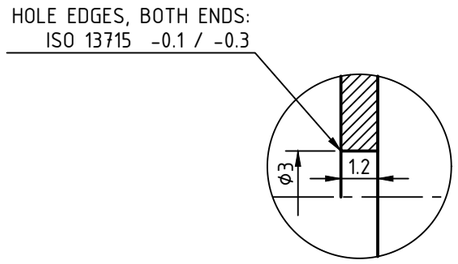
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
Glass phenolic NEMA LI 1 G-3	38.8 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	2:1		mm

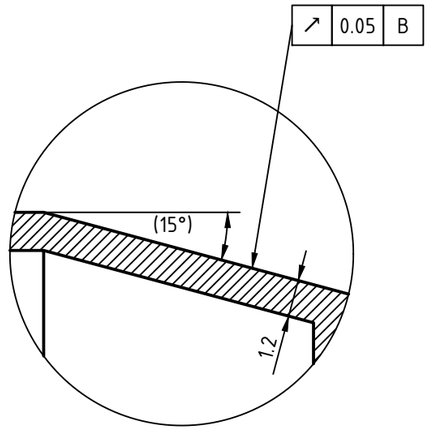
LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	NOZZLE INSULATOR Phenolic washer on the holder face - bore continues the 35° convergent

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-105	A	1 / 1



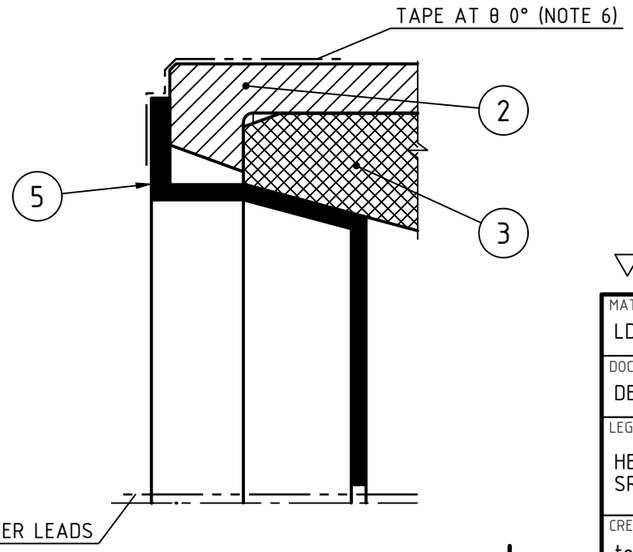


DETAIL D (5:1)



DETAIL C (5:1)

INSTALLATION (REFERENCE) 2:1



NOTES

1. Material: low-density polyethylene PE-LD (ISO 17855), natural, unfilled, unpigmented, no regrind; density about 0.92 g/cm³. Extruded rod ϕ70. Certificate of conformity.
2. Turned from rod, one cap per firing. Op 1: face the aft end, bore ϕ49.42 and the inner cone to the 1.2 end wall, drill ϕ3, turn ϕ66, part off long. Op 2: on a turned mandrel in the bore, turn the cone, ϕ51.9, seat face A and the plug end to length. Sharp high-rake tools, light finishing cuts, no oil. The geometry also suits injection moulding (no undercuts, core drawn aft).
3. Fit: slip fit in the 15° graphite divergent of insert SRM100-104, 0.05 radial clearance at maximum material. All size and station tolerances on the plug go towards clearance: at worst it touches the graphite (profile 0.1) along a line and never wedges; a 15° cone is not self-locking and LDPE cannot score graphite. Seat face A bears on the aft face of holder SRM100-102, but the graphite (and its divergent) is placed by the holder's lip, 6 +0.1/0 (SRM100-102): the clearance holds only with that tolerance. Each 0.1 the lip is thinner moves the divergent 0.027 radially onto the plug; 0.19 thin gives contact. ϕ51.9 clears the lip bore ϕ54 by over 1 mm and the plug clears the graphite's 45° exit break (SRM100-104, stations 6-7).
4. Measure at 20 ± 2 °C (LDPE grows about 0.01 per K on ϕ52). Runouts on the Op 2 mandrel. Inspect 100 %: plug diameters, stations 6 and 16 from A, walls, ϕ3; trial fit in the holder with the insert fitted: enters by hand, flange seats with no rock, lifts out by hand. Visual: no burrs, swarf, cracks or dirt.
5. Edges: both ends of the ϕ3 lead hole ISO 13715 -0.1 / -0.3, no burr (it would cut the lead insulation); other edges per the title block. No marking on the part: bag each cap and label the bag SRM100-106 A.
6. Fitting (SRM100-000 sheet 1 note 4; README, static-fire procedure): fitted at assembly as the debris seal (taped per this note) and kept on through storage and transport. At the stand it comes off only to insert the igniter (leads shunted, pushed through the throat about 615 mm to the middle of the forward grain's core, out through the cap's ϕ3 hole from the plug end), then it is refitted, flange on the holder, and re-taped with new tape. The igniter always goes in last, at the stand: never move a motor with an igniter in it. Tape: two strips of PVC electrical tape, 19 wide, centred at θ 0° and 180°, 90° from the temperature labels at θ 90° and 270° (SRM100-102 note 8), from the flange's aft face over its rim and the holder's aft face onto its aft body ϕ71.58, the ends at most 15 long there; never over a label, the cup opening or the leads.
7. Removal: the cap blows out at ignition (a moisture and debris seal, not a pressure closure). Misfire: disarm and wait at least 60 s (5 min advised for an N motor) before approaching; cut the tape, pull the cap off along the leads, then remove the igniter. Single use: scrap the cap after every firing or misfire; never re-use it. After firing, once the temperature labels have been read and recorded (before extraction, SRM100-102 note 15), remove the tape residue from the holder with IPA.
8. Store bagged, clean and dry, away from solvents, heat and sunlight.
9. Used on SRM100-000 item 5 (nozzle cap: fitted at assembly as the debris seal, refitted at the stand over the igniter leads). Item numbers in the installation view per SRM100-000.

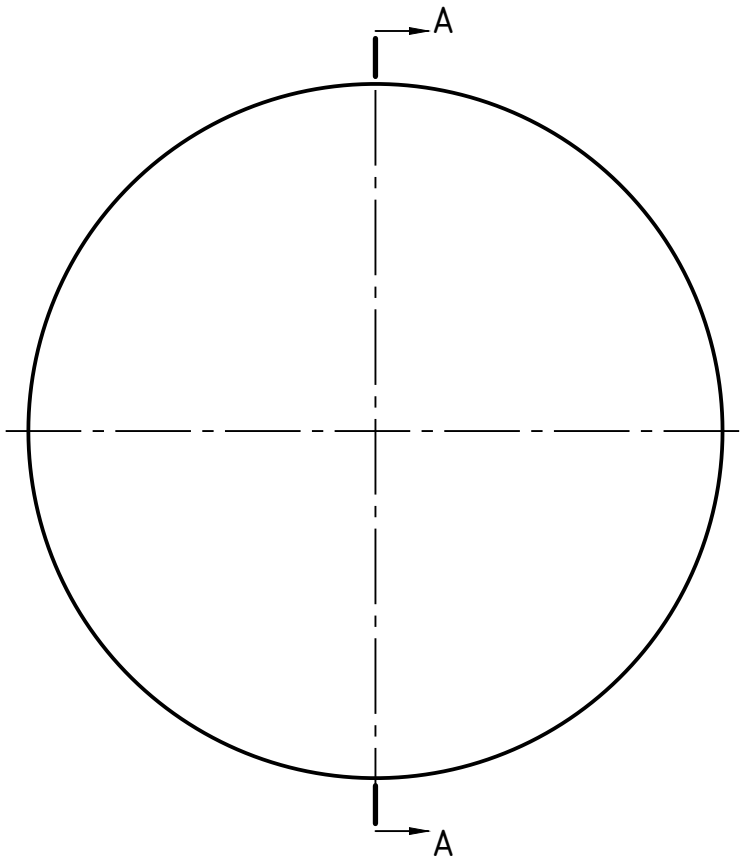
Ra 3.2

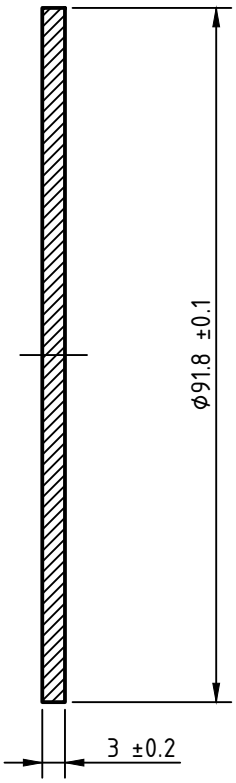
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
LDPE natural, unfilled, rod ϕ70	6.6 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	2:1		mm

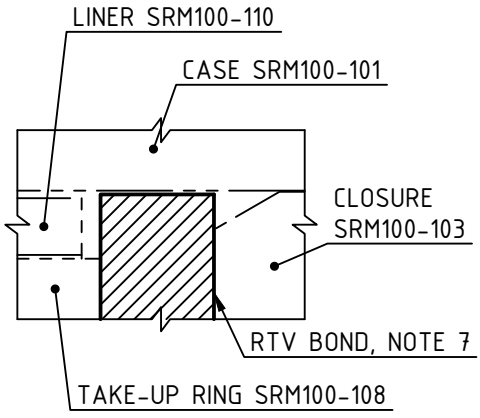
LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	NOZZLE CAP Moisture / debris cap - LDPE plug in the 15° divergent

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-106	A	1 / 1





SECTION A–A



NOTES

1. Material: EPDM, peroxide cured, no fabric insert, to ASTM D2000 M2DA610; certificate required.
2. Stock: sheet 3 ±0.2, smooth faces, no release agent or bloom. The thickness does not stack: the take-up gap is measured at assembly (SRM100-000 sheet 3).
3. Cut to fwd_insulator.dxf (1:1, mm, one OD circle on layer CUT): full disc, no hole; pure water jet or CNC oscillating knife. No laser. Edges square, no tears or nicks.
4. ISO 10579-NR: free state, lying flat, at 23 °C.
5. Inspect 100 %: OD at 2 × 90° by template or caliper, no squeeze; thickness at 4 points, dial gauge, flat foot, light load. No cuts or voids.
6. Fit in the forward seal zone of case SRM100-101, honed to A +0.030/0, A ≤ 92.070: bore ø92.000 … 92.10, diametral clearance 0.10 … 0.40.
7. Bond after the hydrostatic proof to the aft face of closure SRM100-103: abrade both faces P180, IPA wipe, silicone primer SS4004P or equal, RTV106 or equal over the whole face, bondline 0.1 – 0.2; OD centred on the land within 0.1. Cure 24 h, 23 °C.

INSTALLED (5:1) – REFERENCE

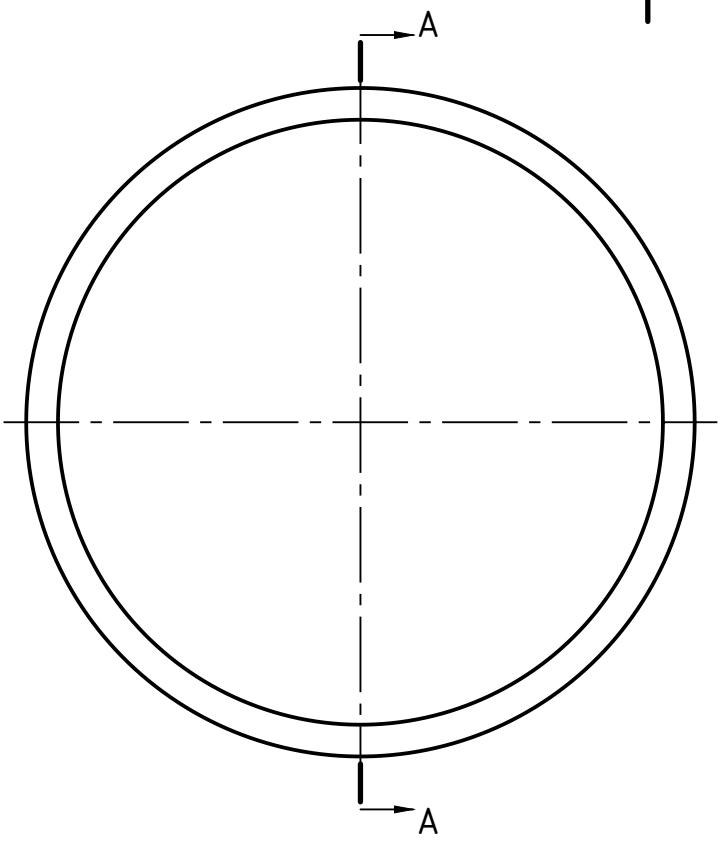
8. At assembly grease the OD edge: Parker Super O-Lube or DC-55, thin film.
9. Replace after every firing. Bag label: SRM100-107, batch, cure date; do not mark the part.
10. USED ON: SRM100-000 motor, item 7 (1 off); not fitted in SRM100-001.

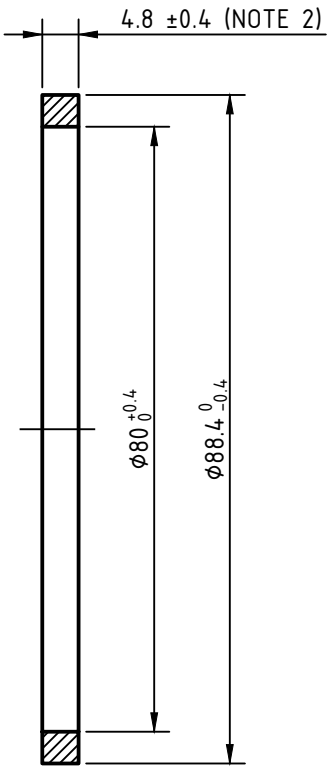
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
EPDM, 60 ±5 Shore A ASTM D2000 M2DA610, sheet 3	22.8 g	ALL TOLERANCED	–	SQUARE, AS CUT

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	FORWARD INSULATOR EPDM disc on the forward closure face – cut from 3 sheet

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-107	A	1 / 1



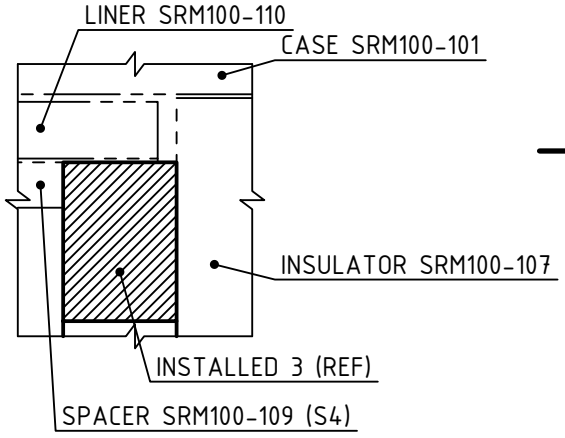


SECTION A-A

FREE STATE

NOTES

1. Material (certified): closed-cell silicone sponge, medium, ASTM D1056 2A2 / AMS 3195, e.g. Rogers BISCO HT-800; free density 0.30 – 0.40 g/cm³; compression set ≤ 10 % (22 h at 100 °C, 50 % deflection).
2. Stock: skinned 3/16 in sheet, free 4.8 ±0.4; where note 6 selects it, 3.2 or 6.4 ±0.4 of the same grade. Do not split or skive.
3. Cut to takeup_ring.dxf (1:1, mm, layer CUT) by steel rule die or oscillating knife; water jet acceptable; no laser or hot wire. The DXF sizes are limits (Ø88.4 max, Ø80 min): aim Ø88.2 / Ø80.2.
4. Inspect 100 %, no tears: free thickness t_s at 4 points 90° apart (dial gauge, flat foot, light load), mean on the bag; OD, ID by template, no squeeze.
5. Fit: OD in liner SRM100-110 bore Ø88.6 +0.1/0, diametral clearance 0.20 … 0.70 (free), whatever the case bore (Ø92.000 … 92.10). Bears on spacer SRM100-109 (S4) and insulator SRM100-107.
6. Take-up (SRM100-000 sheet 3): d_S from the case forward end face to the top of S4 and t from the closure flange face to the insulator's aft face, 4 points each, mean; g = d_S − t (nominal 3.00 ±0.50). Accept 0.50 ≤ g / t_s ≤ 0.75 (25-50 % compression), no shims: sheet 3.2 for g 1.60-2.40, 4.8 for g 2.40-3.60, 6.4 for g 3.20-4.80. Lay the ring on S4.



INSTALLED (5:1) – REFERENCE

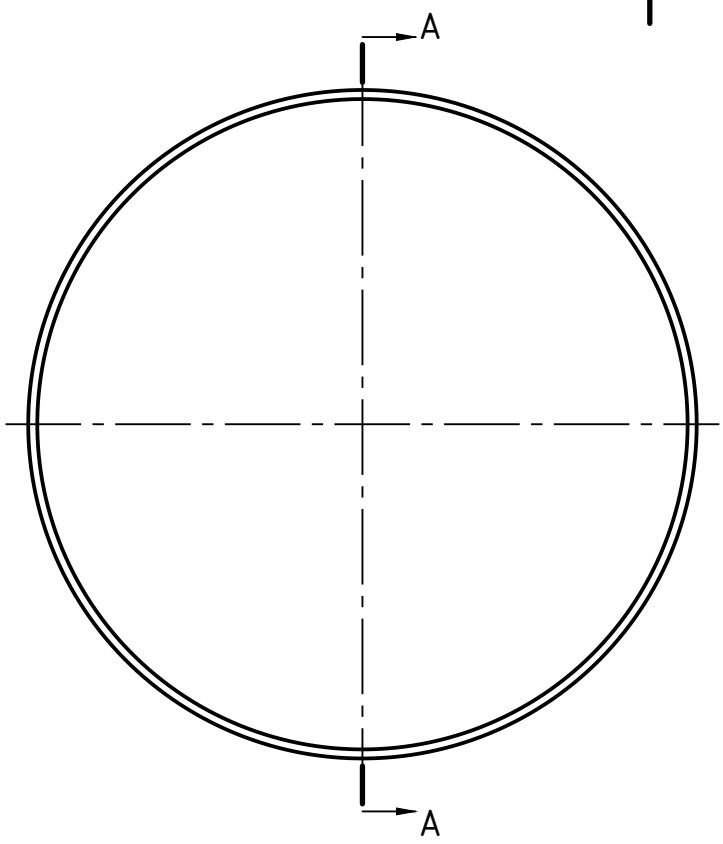
7. ISO 10579-NR: free state, lying flat, at 23 °C. The 3D model and title-block mass are the ring installed, 3 thick: 3.33 cm³ × 0.50 g/cm³. As cut: 5.33 cm³ × 0.30 – 0.40 g/cm³ = 1.6 – 2.1 g.
8. Replace after every firing. Bag label: SRM100-108, batch, sheet, t_s; do not mark the part.
9. USED ON: SRM100-000 motor, item 12 (1 off); not fitted in SRM100-001.

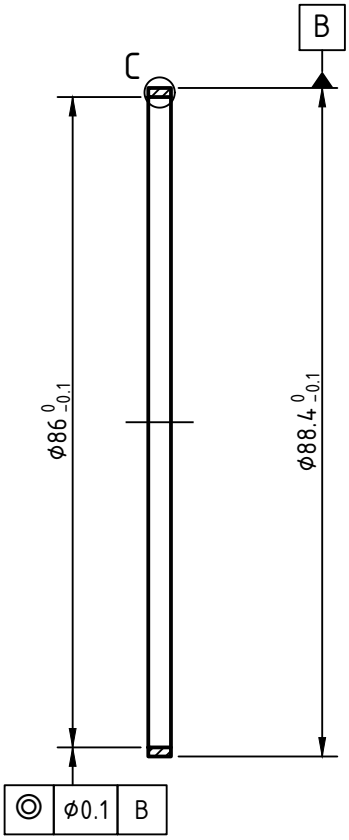
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
Silicone sponge, closed cell ASTM D1056 2A2, sheet 4.8	1.7 g	ALL TOLERANCED	–	SQUARE, AS CUT

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	TAKE-UP RING Grain-stack axial take-up – silicone sponge, cut from 4.8 sheet

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-108	A	1 / 1

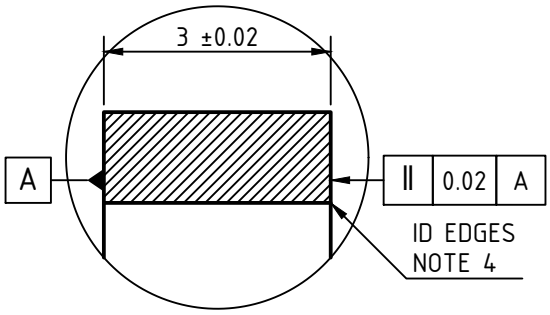




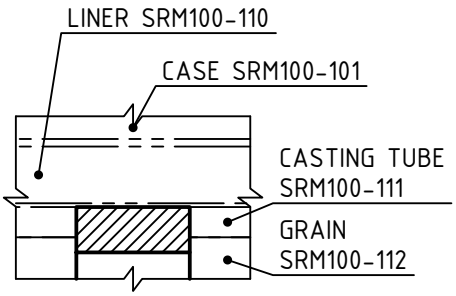
SECTION A-A

NOTES

1. Material (certified): linen phenolic, NEMA LI 1 grade LE (IEC 60893-3-4 PF CC 203 or equal).
2. Stock: sheet 4 (5/32 in), surface-ground equally on both faces to 3 ±0.02 before profiling. Machine dry, no coolant.
3. Cut to grain_spacer.dxf (1:1, mm, layer CUT) by CNC router (carbide compression cutter) or turn a clamped stack on a mandrel; no laser, water jet or punching. The DXF sizes are upper limits: aim ϕ88.35 / ϕ85.95.
4. Edges: ISO 13715 -0.2, but both ID edges burr-free, -0.05 max: the ID band bears on the grain face.
5. Fit: OD in liner SRM100-110 bore ϕ88.6 +0.1/0, diametral clearance 0.20 … 0.40, whatever the case bore (ϕ92.000 … 92.10). The ID covers 0.40 … 0.45 of each grain face (ϕ86.8, radial, concentric).
6. Inspect 100 %: t at 4 points (micrometer), mean on the bag label; OD and ID at 2 × 90°; parallelism on a surface plate, face A down.
7. Replace after every firing. Bag label SRM100-109, batch, mean t; store sealed, with desiccant.
8. USED ON: SRM100-000 motor, item 11 (5 off); not fitted in SRM100-001.



C (10:1)



INSTALLED (5:1) - REFERENCE

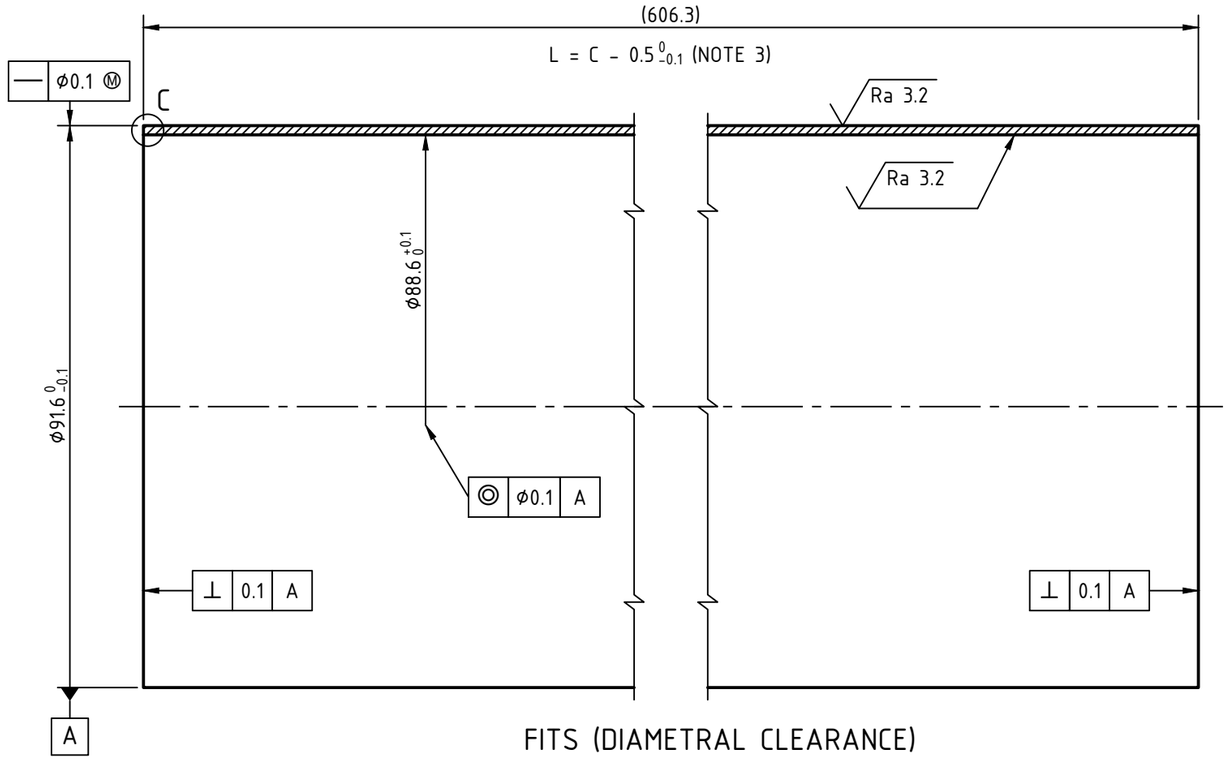
Ra 3.2

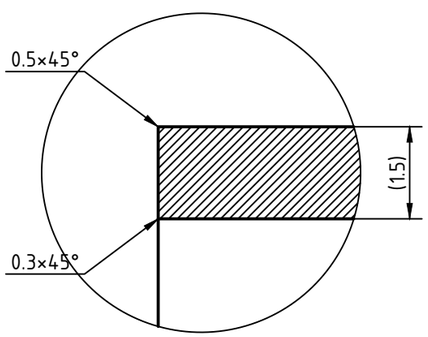
MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
Linen phenolic, NEMA LI 1 LE sheet 4, faces ground to 3	1.3 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.2, NOTE 4

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	GRAIN SPACER Grain segment spacer - linen phenolic, 5 off per motor

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-109	A	1 / 1





C (10:1)

FITS (DIAMETRAL CLEARANCE)

MATING PART	BORE	SHAFT	CLEARANCE
Liner OD in case bore SRM100-101, full length	⌀92.000 … 92.10	⌀91.50 … 91.60	0.40 … 0.60
Casting tube SRM100-111 in liner bore	⌀88.60 … 88.70	⌀88.30 … 88.40	0.20 … 0.40
Grain spacer SRM100-109 in liner bore	⌀88.60 … 88.70	⌀88.30 … 88.40	0.20 … 0.40
Take-up ring SRM100-108 in liner bore	⌀88.60 … 88.70	⌀88.00 … 88.40	0.20 … 0.70

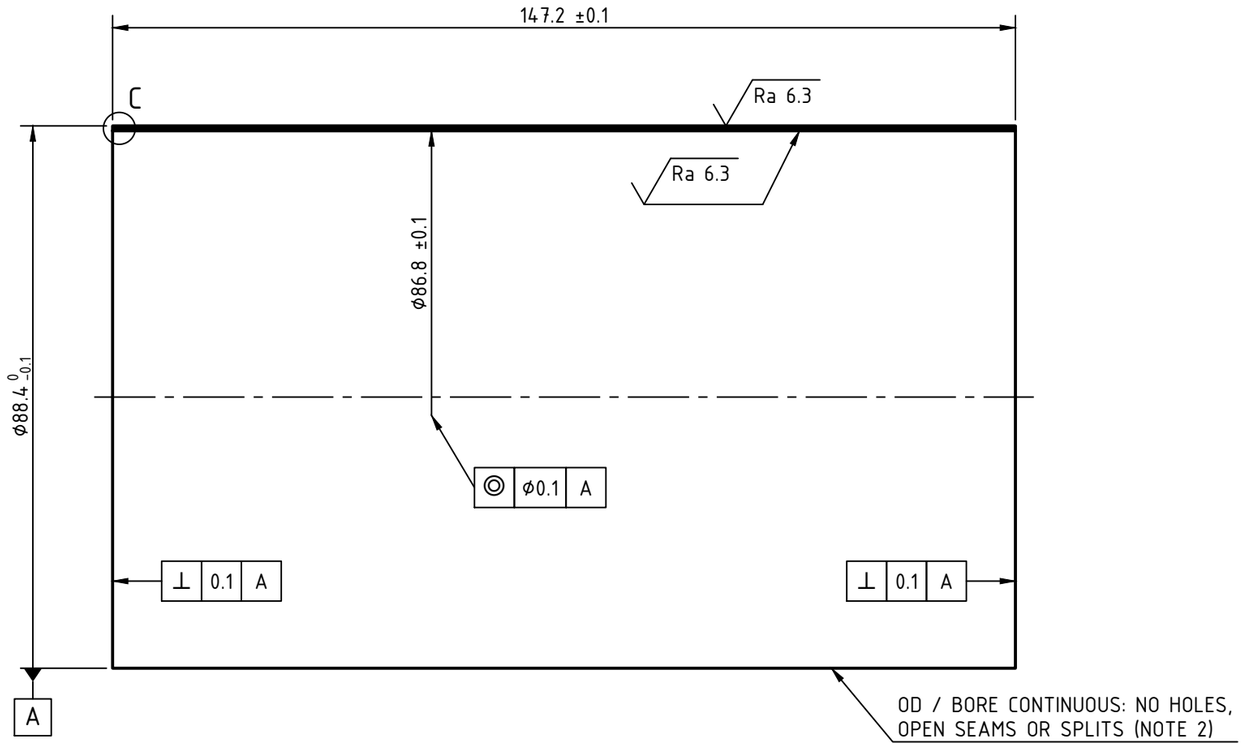
Both liner ends lie in the honed seal zones of SRM100-101 (45 +5/0 from each case end face: overlap ≥ 2 aft, ≥ 3.5 fwd), held to A +0.030/0 with 92.000 ≤ A ≤ 92.070, A recorded per zone: the bore is ≤ 92.10 everywhere, zones included, so the clearance 0.40 … 0.60 holds over the full length. Straightness ⌀0.1 Ⓜ: liner virtual size ⌀91.70 = smallest straight case bore, GO plug ⌀91.70 (bore min 92.000, surface lines straight within 0.3, SRM100-101).

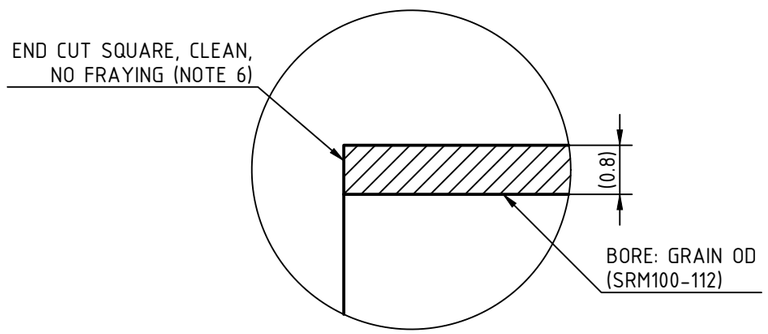
NOTES

1. Material: phenolic-impregnated kraft paper tube, convolute or spiral wound; paper-base phenolic to NEMA LI 1 grade X / XX or equal, density approx. 1.35 g/cm³. Stock: tube bought to the drawn OD and bore, or OD ground / bore honed from an oversize tube; supply length ≥ 612. Free of delamination, blisters, soft spots and cracks. Certificate of conformity required.
2. Datum A is the OD axis. Sizes are two-point sizes at 20 °C on a tube conditioned dry (sealed with desiccant ≥ 24 h): paper phenolic swells with humidity. Store sealed.
3. Length L is cut at assembly to the measured cavity C, nozzle-insulator forward face to forward-insulator aft face (SRM100-000): L = C − 0.5 0/−0.1. (606.3) is L at the nominal C = 606.8. The liner carries no axial load: before the forward closure goes in, the depth from the case forward end face to the liner forward end must be ≥ t + 0.30 at 4 points, t = closure flange bearing face to forward-insulator aft face (SRM100-000 sheet 3, note 8).
4. Edges drawn sharp. Break both OD end edges 0.5x45° (protects the honed seal zones of SRM100-101 when the liner is pushed in) and both bore end edges 0.3x45°. Cut ends clean: no loose fibres, no crushed or split ends.
5. Clean and dry inside and out before assembly. Grease the OD at assembly: Parker Super O-Lube or DC-55, thin film; no grease in the bore. Must slide by hand through case SRM100-101 over its full length. Casting tubes, spacers and take-up ring must slide freely in the bore.
6. Inspect 100 %: OD and bore at both ends and mid-length, 2 directions 90° apart; wall ≥ 1.35 at 4 positions each end; straightness ⌀0.1 Ⓜ: a full-length sleeve gauge ⌀91.70 passes over the OD by hand; length at 4 positions 90° apart after cutting. Visual inside and out. Record with the case serial.
7. Mark SRM100-110, batch and the serial of the case it is cut for on the OD, within 30 of one end: indelible ink, 3 mm characters. Do not engrave or stamp.
8. Installed from motor z 43 (bears on nozzle insulator SRM100-105) to z 649.3; the forward end floats 0.5 short of forward insulator SRM100-107.
9. Used on SRM100-000 item 8 (1 off). Single use: replaced after every firing (README, After every firing).

Ra 6.3 (√)

MATERIAL: Paper-base phenolic tube NEMA LI 1 grade X / XX or equal	MASS: 347.5 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: Ra 6.3	EDGES: ISO 13715 −0.3 / +0.3
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: LINER — Case liner / thermal barrier, ⌀91.6 / ⌀88.6 - paper phenolic			
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-110	REV: A / SHEET: 1 / 1





C (10:1)

NOTES

1. Material: spiral-wound kraft paper tube, plain kraft (not phenolic-impregnated), wall 0.8, density approx. 0.95 g/cm³. Stock: tube bought to the drawn bore and wall, cut to length. Plies bonded over their full area. Certificate of conformity required.
2. Propellant grain SRM100-112 is cast into this tube and stays in it: the tube is the grain's OD inhibitor. It must be continuous: no holes, open or overlapping seams, splits, blisters, delamination, dents or crushed ends. Inspect against a light.
3. Bore clean and dry: free of oil, wax, dust and loose fibres. OD free of raised seams and adhesive lumps.
4. Datum A is the OD axis. Sizes are two-point sizes at 20 °C on a tube conditioned dry (sealed with desiccant ≥ 24 h): kraft swells with humidity. Store sealed.
5. Length = grain length. In the finished segment the tube ends are flush with the grain end faces within 0.1 (SRM100-112).
6. Edges drawn sharp: ends cut square and clean, edge break 0.2 max, no burr or fraying. The ends bear on the grain spacers.
7. OD, also after the grain is cast in it, must pass freely over its full length through a ø88.60 ring gauge (liner SRM100-110 bore min).
8. Inspect 100 %: OD and bore at both ends, 2 directions 90° apart; length at 4 positions 90° apart; visual inside and out. Record per batch.
9. Mark SRM100-111 and batch on the OD within 20 of one end: indelible ink, 3 mm characters. The grain marking (SRM100-112) goes on the same OD.
10. Used on SRM100-000 item 9 (4 off), one per grain SRM100-112 (item 10). Single use: replaced with its grain after every firing (README, After every firing).

FIT AND POSITION

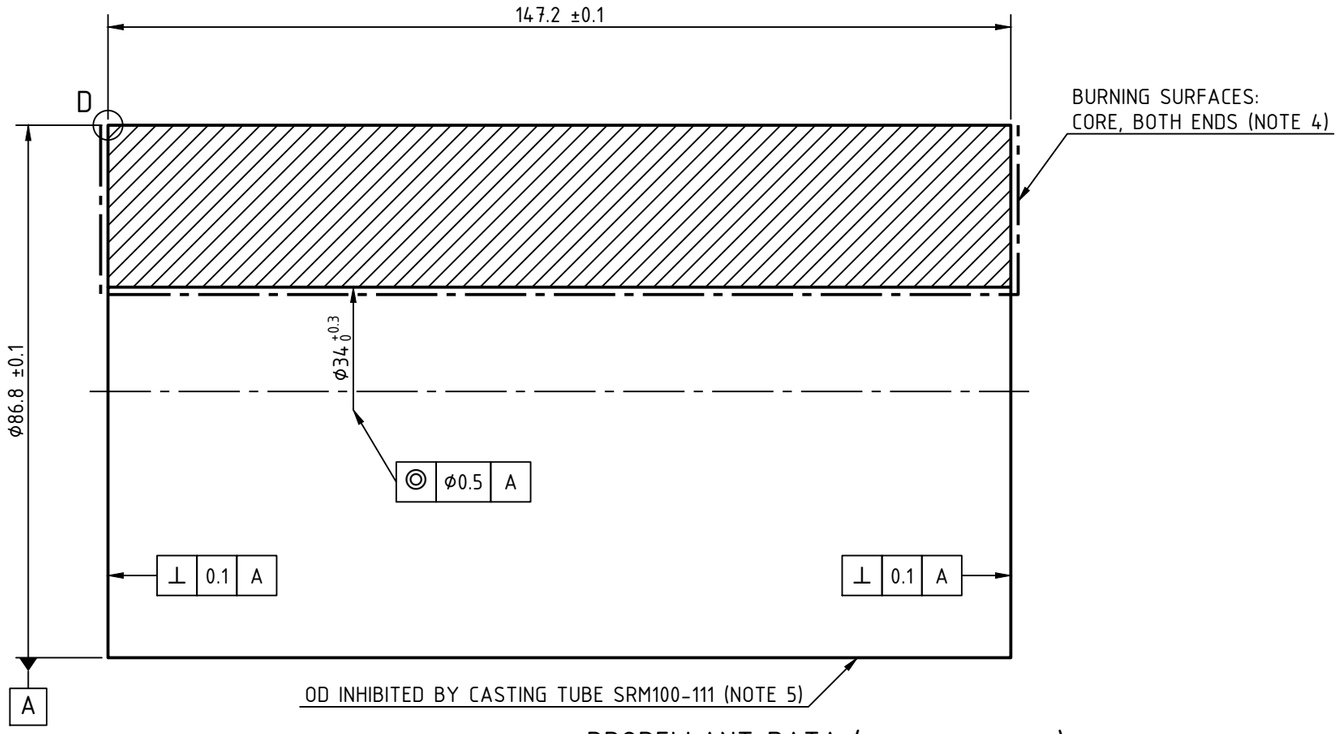
ITEM	VALUE
Liner SRM100-110 bore	ø88.60 … 88.70
Tube OD in liner bore	clearance 0.20 … 0.40 (diametral)
Case bore SRM100-101	no contact: the liner wall is between
Bore = grain SRM100-112 OD	ø86.8 ±0.1, cast in place
Aft face, motor z (tubes 1 … 4)	46 / 196.2 / 346.4 / 496.6
Ends bear on	grain spacers SRM100-109

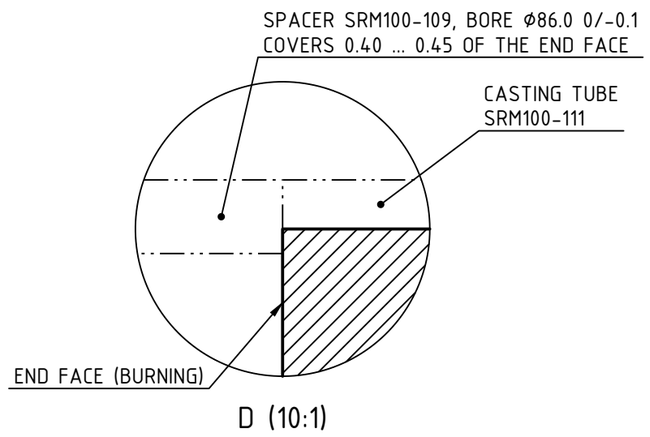
Ra 6.3 (√)

MATERIAL: Spiral-wound kraft paper tube stock ID 86.8 x 0.8 wall	MASS: 30.8 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: Ra 6.3	EDGES: ISO 13715 -0.2	
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm	
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: CASTING TUBE — Grain casting tube / OD inhibitor, ø88.4 / ø86.8 x 147.2 - kraft				
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-111	REV: A	SHEET: 1 / 1

NOTICE – PROPELLANT MANUFACTURE

Propellant grains are made only by a person or facility licensed / certified for propellant manufacture under the applicable local law. This drawing defines geometry, properties and acceptance only; it is not a manufacturing instruction and gives no mixing, casting or curing process.





D (10:1)

PROPELLANT DATA (srm_params.py)

PROPERTY	VALUE
Propellant	APCP 70/16/14 (AP/Al/HTPB), mass %
Density, as cast	1718 kg/m³ = 98 % of TMD 1753: nominal
Density, paper	1770 kg/m³ (above TMD): pressure / hardware bound
Burn rate, 20 °C	$r = a \cdot P^n$, P in MPa: a = 3.5 mm/s (at 1 MPa), n = 0.35
Temp. sens.	σp = 0.2 %/K; grain ≤ 40 °C fired
c*	1550 m/s theoretical; efficiency 0.95
Heat ratio γ	1.21

a, n: the paper's generic values; measure on the batch (NOTE 2).

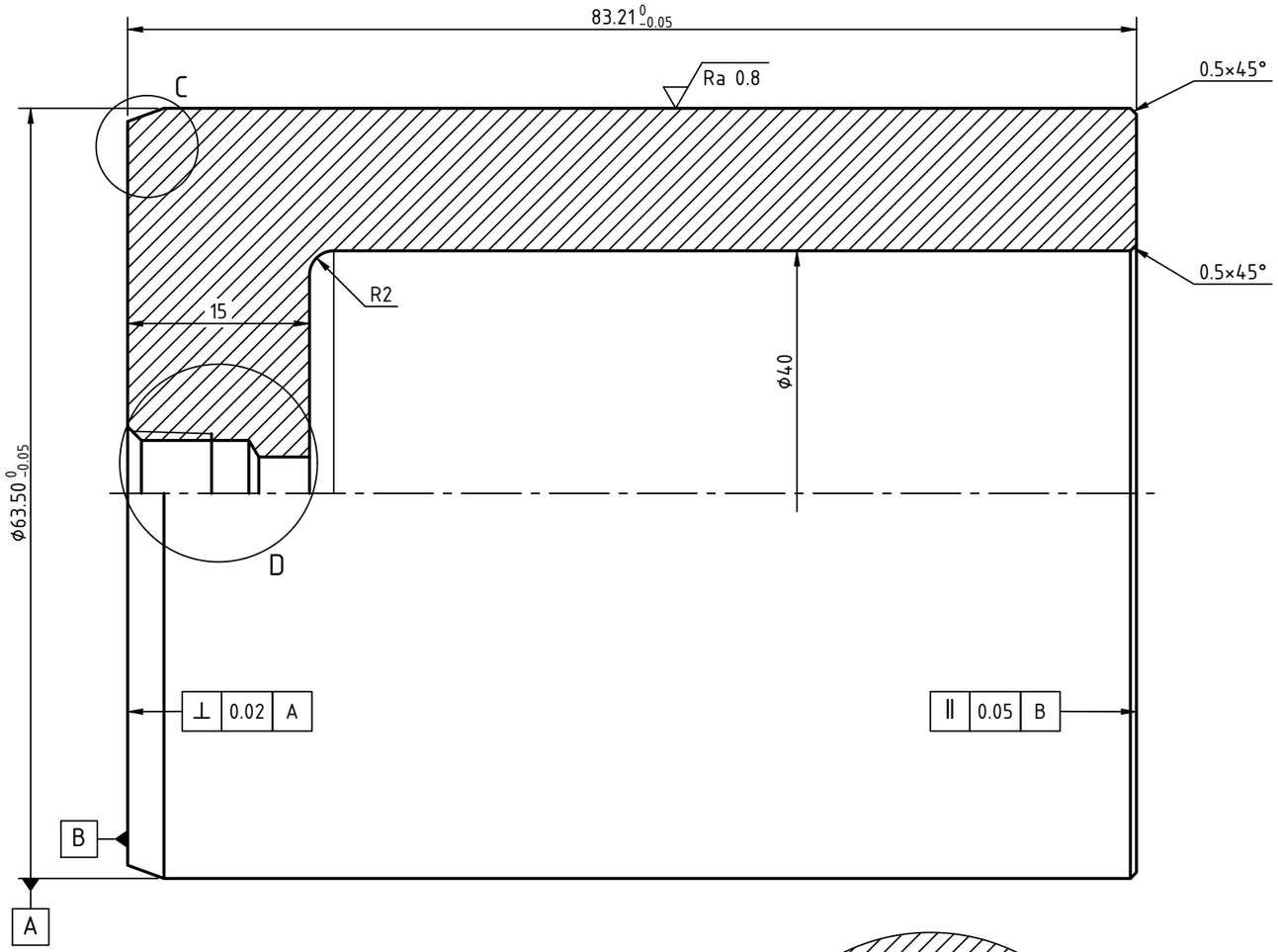
NOTES

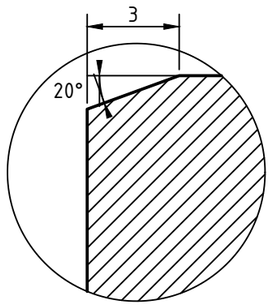
1. Material: composite propellant APCP 70/16/14, ammonium perchlorate / aluminium / HTPB binder in % by mass, cast into casting tube SRM100-111, which stays bonded to the grain as its OD inhibitor. Properties: PROPELLANT DATA table.
2. Burn rate: a and n are the paper's generic values, not a batch property. Before loading, the burn rate of the actual batch is measured (strand burner or sub-scale motor), entered in srm_params.py and both simulations are re-run. A batch 21 % faster than the table reaches MEOP 7.0 MPa at 40 °C (README, Burn-rate sensitivity: Lenoir-Robillard erosion, bound inputs).
3. Datum A is the OD axis. The OD is the bore of casting tube SRM100-111 (ø86.8 ±0.1, coaxial ø0.1 to the tube OD); it is not measured on the grain: simulate A on the tube OD. Sizes at 20 °C.
4. Burning surfaces (thick chain line): core and both end faces, uninhibited: no inhibitor, tape, adhesive, release agent or tube overlap on them. Each end face flush with the tube end within 0.1.
5. OD inhibited by the casting tube: the grain is bonded to the tube over its full area; no separation visible at either end, no propellant on the tube OD.
6. Mass, net propellant (segment minus its tube): 1266.8 g ±1.5 % = 1247.8 … 1285.8 g (bulk density 1692 … 1744 kg/m³). Matched set: each of the 4 grains of a motor within 1 % of their mean. PROVISIONAL: this tolerance is an assumption, not in the design basis.
7. Visual, 100 %, core and both end faces: no cracks of any size, no voids or pits over 1 mm, no foreign matter, no unmixed or discoloured areas; edge chips within ISO 13715 -0.5. A grain with a crack is rejected. Surfaces as cast: no roughness value applies.
8. Dimensional, 100 %: core at both ends, 2 directions 90° apart; web (OD to core) at 4 positions each end, spread ≤ 0.5; length at 4 positions 90° apart. Record per grain with the mass and batch.
9. One batch per motor: the 4 grains and the burn-rate sample of note 2 come from the same batch.
10. Mark on the casting tube OD, never on the propellant: SRM100-112, batch, grain no., cast date, net mass in g. Indelible ink, 3 mm characters. Hazard labelling, storage and transport per local law.
11. Installed with the aft face at motor z 46 / 196.2 / 346.4 / 496.6 (grain 1 aft), between grain spacers SRM100-109. Either end may face aft: the segment is symmetric. The casting mandrel is tooling and is not covered by this drawing.
12. Used on SRM100-000 item 10 (4 off). Single use: consumed in the firing; a new matched set of 4 (note 6) for every firing (README, After every firing).

GRAIN DATA (REFERENCE, NOMINAL GEOMETRY)

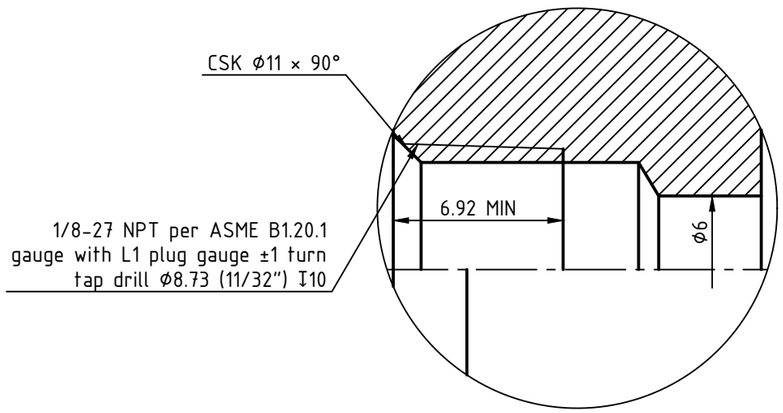
ITEM	VALUE
Volume	737.4 cm³ per grain
Initial burning area	25 742 mm² per grain (core + 2 ends)
Of which under spacers	217 … 244 mm² = 0.84 … 0.95 % (detail D), not deducted
Web; port / throat area	26.4; 2.19 (throat ø23)
Kn, 4 grains, geometric	247.8 initial, 279.5 max (uniform regression, srm_params.kn)
Kn, 4 grains, simulated	281.1 peak, nominal run (erosive burning, srm_ballistics)
Propellant, 4 grains	5.067 kg as cast; 5.221 kg at 1770

MATERIAL: APCP 70/16/14 (AP / Al / HTPB) cast in casting tube SRM100-111	MASS: 1266.8 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: as cast	EDGES: ISO 13715 -0.5
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: PROPELLANT GRAIN — BATES segment ø86.8 / ø34 x 147.2 - APCP 70/16/14, cast in SRM100-111			
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-112	REV: A / SHEET: 1 / 1





C (5:1)



D (4:1)

FILL / BLEED PORT (the motor's only port)

NOTES

1. Material: AISI 304 / 1.4301 austenitic stainless, annealed, round bar to ASTM A479. Material certificate EN 10204 3.1.
2. PROOF ONLY: never fired, never fitted for a firing. It takes the graphite insert SRM100-104's place in the hydrostatic proof SRM100-001, so the holder's lip, its bore gland and all five O-rings are proofed while the graphite stays dry.
3. Carries the insert's fits. OD ϕ63.50 0/-0.05: slip fit in the SRM100-102 bore ϕ63.58 +0.025/0, slides in by hand, aft end first; do not press. Length 83.21 0/-0.05 against the seat depth 83.21 +0.05/0: forward face flush to 0.10 below the holder face.
4. Face B bears on the holder's lip from ϕ54 out to ϕ61.32 (3.66 radial). Closed, the dummy puts 33 kN on the lip at 10.5 MPa, against the insert's 19 kN at MEOP 7 MPa.
5. OD Ra 0.8, free of axial scratches: the AS568-230 O-ring runs on it. Blend the lead-in into the OD, burr-free: the O-ring passes over it on assembly.
6. Port (detail D): the proof's fill and bleed port, the motor's only port (item 6 has none), on the axis of face B, reached through the holder's lip bore; the ϕ6 passage takes the ϕ4 dip tube (SRM100-001). 1/8-27 NPT per ASME B1.20.1: 90° countersink ϕ11, tap drill ϕ8.73 (11/32") I10, 118° point, then the passage into the cup. Full thread 6.92 min (L1 + L3): taper pipe tap with a 2 … 3 thread chamfer. Rig tee: Loctite 567 or equal on the male thread, first 2 threads clear; no PTFE tape.
7. Deburr, clean and degrease; passivate to ASTM A967. No chips or oil left in the cup, passage or port.
8. Inspect 100 %: OD at three axial positions, length, perpendicularity to A, port with a 1/8-27 NPT L1 plug gauge ±1 turn. Record the values against the serial.
9. Mark SRM100-120, serial and PROOF ONLY on the forward face, between ϕ44 and ϕ60, by laser or dot-peen, 2 mm characters, 0.05 deep max.
10. Extract as the insert: holder out, the insert push sleeve (tooling) through the lip bore onto face B. Never lever or knock.
11. USED ON: SRM100-001 hydrostatic proof assembly, item 17, in place of the insert SRM100-104 (item 3).

Ra 3.2 (✓)

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
AISI 304 / 1.4301, annealed round bar to ASTM A479	1414.1 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	2:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	PROOF DUMMY Hydrostatic-proof stand-in for insert SRM100-104 - 304 stainless

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-120	A	1 / 1